

SELF-SUPERVISED CONTRASTIVE REPRESENTATION LEARNING FOR RARE ASTRONOMICAL TRANSIENT CLASSIFICATION USING MULTI-BAND LIGHT-CURVE SEQUENCES

Research Report

*Submitted in Partial Fulfillment of the Requirements for
B.Sc (Hons) Degree in Information Technology
Specializing in Data Science*

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DECLARATION

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List of Abbreviations

LIST OF ABBREVIATIONS

Abbreviation	Full Form
AI	Artificial Intelligence
ML	Machine Learning
DL	Deep Learning
CNN	Convolutional Neural Network
RNN	Recurrent Neural Network
GRU	Gated Recurrent Unit
LSTM	Long Short-Term Memory
MLP	Multilayer Perceptron
SSL	Self-Supervised Learning
SimCLR	Simple Framework for Contrastive Learning of Representations

Abbreviation	Full Form
BYOL	Bootstrap Your Own Latent
MoCo	Momentum Contrast
NT-Xent	Normalized Temperature-scaled Cross-Entropy Loss
UMAP	Uniform Manifold Approximation and Projection
ZTF	Zwicky Transient Facility
LSST	Legacy Survey of Space and Time
SN Ia	Type Ia Supernova
SN II	Type II Supernova
AGN	Active Galactic Nuclei
QSO	Quasi-Stellar Object
RR Lyrae	RR Lyrae Variable Star
Cepheid	Cepheid Variable Star
EB	Eclipsing Binary
LPV	Long-Period Variable

Abbreviation	Full Form
CV	Cataclysmic Variable
GPU	Graphics Processing Unit
CPU	Central Processing Unit

Abstract

Rare astronomical transients such as kilonovae, tidal disruption events, and cataclysmic variables represent some of the most extreme and scientifically significant astrophysical phenomena observable in the universe. These events offer critical insights into stellar evolution, gravitational wave counterparts, and multi-messenger astronomy. However, the automated classification of such transients remains a formidable challenge due to three primary difficulties: the scarcity of labeled training data for rare event classes, significant class imbalance where common transients dominate the dataset, and the inherently noisy, irregular, and incomplete nature of observational light curves.

This report presents a self-supervised contrastive learning-based framework for multi-band light curve classification using the SimCLR paradigm. The proposed framework effectively exploits large quantities of unlabeled photometric time series data through domain-aware data augmentation strategies including temporal jittering, magnitude scaling, Gaussian noise perturbation, and subsequence cropping. A novel hybrid temporal encoder architecture combining one-dimensional convolutional neural layers and gated recurrent units (GRUs) is employed to capture both local morphological features and global temporal dependencies in the light curve sequences.

The framework is subsequently fine-tuned using a limited number of labeled samples to train a classification head for multi-class astronomical transient identification. Experimental evaluation demonstrates that the proposed contrastive learning framework achieves 83% classification accuracy, substantially outperforming a state-of-the-art autoencoder-based baseline which attains only 68% accuracy. These results validate the superiority of discriminative contrastive representation learning over reconstruction-based self-supervised approaches for rare astronomical transient classification under realistic label-scarce conditions.

Keywords: *astronomical transients, contrastive learning, light curves, rare event classification, self-supervised learning, SimCLR, time-series representation learning, GRU, CNN, data augmentation*

1. Introduction

1.1 Background and Motivation

Time-domain astronomy has undergone a transformative evolution over the past decade, driven by the rapid advancement of large-scale synoptic sky surveys and high-throughput observational technologies. Unlike traditional astronomical studies that relied on static observations of celestial objects, modern time-domain astronomy focuses on capturing the dynamic and transient nature of the universe by repeatedly scanning vast regions of the sky over time. This paradigm shift has resulted in an unprecedented increase in the volume, velocity, and variety of photometric data generated on a daily basis.

State-of-the-art survey facilities such as the Zwicky Transient Facility (ZTF), Panoramic Survey Telescope and Rapid Response System (Pan-STARRS), and the upcoming Vera C. Rubin Observatory’s Legacy Survey of Space and Time (LSST) are capable of producing millions of transient alerts per night. These alerts correspond to astrophysical events that exhibit temporal variability, including supernova explosions, variable stars, active galactic nuclei, and other transient phenomena. The sheer scale of these data streams has rendered manual inspection and classification infeasible, thereby necessitating the development of automated, intelligent classification systems capable of operating in real time.

Within this observational framework, transient events are commonly represented as light curves, which are time-series sequences capturing the variation of an object’s brightness (magnitude) across one or more photometric bands (e.g., g-band, r-band). These light curves encode rich temporal and spectral information about the underlying astrophysical processes, including rise times, decay patterns, periodicity, and irregular fluctuations. Consequently, the analysis and classification of light curves have become central tasks in modern computational astronomy.

Among the wide spectrum of transient phenomena, rare astronomical transients occupy a position of exceptional scientific significance. These include events such as:

- Kilonovae, which arise from neutron star mergers and are key to understanding heavy element nucleosynthesis through rapid neutron capture (r-process),
- Tidal Disruption Events (TDEs), where stars are gravitationally disrupted by supermassive black holes, providing insights into black hole accretion physics,
- Cataclysmic Variables (CVs), which are binary star systems exhibiting sudden and dramatic brightness variations due to mass transfer processes.

These rare events are invaluable for probing extreme physical environments, testing fundamental theories of gravity and high-energy astrophysics, and contributing to the emerging field of multi-messenger astronomy, where electromagnetic observations are combined with gravitational wave detections.

However, despite their scientific importance, the identification and classification of rare transients present significant challenges. The primary difficulty stems from their intrinsic rarity, which leads to highly imbalanced datasets where common events (e.g., Type Ia supernovae) dominate the observational distribution. In many cases, rare transient classes may constitute less than 1% of the total dataset, making it extremely difficult for conventional machine learning models to learn meaningful patterns for these minority classes.

Furthermore, the process of obtaining labeled data for rare events is both time-consuming and resource-intensive. Accurate labeling often requires spectroscopic follow-up observations using high-resolution telescopes, as well as expert analysis by astronomers. As a result, labeled datasets for rare transients are typically small, incomplete, and biased, limiting the effectiveness of traditional supervised learning approaches that rely heavily on large volumes of annotated data.

In addition to label scarcity and class imbalance, the quality of observational data introduces further complexity. Photometric light curves are often characterized by:

- Irregular temporal sampling, due to varying observational cadences,
- Missing data points, caused by weather conditions or instrument limitations,
- Measurement noise, arising from atmospheric interference and detector imperfections,
- Heterogeneous sequence lengths, reflecting incomplete or partial observations.

These factors collectively produce highly noisy and non-uniform time-series data, which pose significant challenges for standard machine learning models that assume clean, regularly sampled inputs.

Given these limitations, there is a growing need for robust, data-efficient, and noise-tolerant learning frameworks that can effectively leverage the abundance of unlabeled observational data while minimizing reliance on scarce labeled examples. In this context, self-supervised learning, and in particular contrastive representation learning, has emerged as a promising paradigm. By learning to extract meaningful representations from unlabeled data through carefully designed pretext tasks, these methods enable models to capture the underlying structure of the data without explicit supervision.

This research is motivated by the hypothesis that contrastive self-supervised learning can address the fundamental challenges of rare transient classification by:

- Exploiting large volumes of unlabeled light curve data,
- Learning discriminative and invariant representations robust to noise and irregularity,
- Improving generalization performance under limited labeled data conditions.

By integrating domain-specific augmentation strategies with advanced temporal modeling architectures, this work aims to contribute toward the development of scalable and intelligent classification systems for next-generation astronomical surveys.

1.2 Problem Statement

The automated classification of rare astronomical transients constitutes a complex and multi-dimensional problem, arising from the interplay of data limitations, observational constraints, and intrinsic characteristics of astrophysical phenomena. Despite significant advancements in machine learning and data-driven astronomy, the accurate identification of rare transient events remains an open challenge due to three fundamental and interconnected issues: label scarcity, class imbalance, and data irregularity.

1.2.1 Label Scarcity

One of the most critical challenges in astronomical transient classification is the extreme scarcity of labeled data, particularly for rare event classes. Unlike conventional machine learning domains where labeled datasets can be curated at scale, the labeling process in astronomy is highly dependent on spectroscopic confirmation, which requires access to expensive telescope resources and expert interpretation.

Spectroscopic follow-up observations are time-intensive and cannot feasibly be performed for every detected transient event. As a result, only a small fraction of observed transients are assigned reliable class labels. This leads to datasets where the number of labeled instances per rare class is extremely limited, often consisting of only a few confirmed samples.

From a machine learning perspective, this scarcity can be formally described as a scenario where the labeled dataset

$$\mathcal{D}_l = \{(x_i, y_i)\}_{i=1}^{N_l}$$

is significantly smaller than the unlabeled dataset

$$\mathcal{D}_u = \{x_j\}_{j=1}^{N_u}, \text{ with } N_l \ll N_u.$$

Under such conditions, traditional supervised learning algorithms are prone to overfitting, as they attempt to learn complex decision boundaries from insufficient

training examples. This results in poor generalization performance, particularly for rare classes that lack adequate representation in the training data.

1.2.2 Class Imbalance

In addition to label scarcity, astronomical datasets are characterized by severe class imbalance, where the distribution of observed transient events is highly skewed toward a few dominant classes. Common phenomena such as Type Ia supernovae and core-collapse supernovae constitute the majority of observations, while rare events such as kilonovae or tidal disruption events may account for less than 1% of the total dataset.

This imbalance can be represented as a probability distribution over classes:

$$P(y = c_k) \gg P(y = c_r), \text{ for common class } c_k \text{ and rare class } c_r.$$

Such skewed distributions introduce a strong bias in conventional classifiers, which tend to optimize for overall accuracy by favoring majority classes. Consequently, the model may achieve high aggregate performance while systematically failing to detect rare but scientifically important events.

This issue is particularly problematic in astronomy, where false negatives for rare events are significantly more critical than false positives, as missing a rare transient may result in the permanent loss of valuable scientific information. Therefore, classification models must be designed not only for accuracy but also for robust detection of minority classes, emphasizing metrics such as recall and F1-score.

1.2.3 Data Quality and Irregularity

A further layer of complexity arises from the inherent nature of photometric observational data, which is often noisy, incomplete, and irregularly sampled. Unlike standard benchmark time-series datasets that assume uniform sampling intervals, astronomical light curves exhibit highly variable temporal structures due to observational constraints.

Key challenges include:

- **Irregular Sampling:** Observations occur at non-uniform time intervals, influenced by survey scheduling, weather conditions, and telescope availability.
- **Missing Data:** Significant portions of the light curve may be absent due to observational gaps or detection limits.

- **Measurement Noise:** Photometric measurements are affected by atmospheric turbulence, detector sensitivity, and background noise.
- **Heterogeneous Sequence Lengths:** Different transient events are observed over varying durations, leading to inconsistent sequence lengths.

Formally, each light curve can be represented as a multivariate time series:

$$x_i = \{(t_1, f_1), (t_2, f_2), \dots, (t_T, f_T)\},$$

where the time steps t_k are irregularly spaced, and the observations f_k may contain missing or noisy values.

These characteristics violate the assumptions of many traditional machine learning models, which expect clean, regularly sampled inputs. As a result, standard approaches such as CNNs and RNNs struggle to effectively capture meaningful temporal patterns under such conditions.

1.2.4 Limitations of Traditional Supervised Approaches

Traditional supervised learning methods, including random forests, convolutional neural networks (CNNs), and recurrent neural networks (RNNs), have demonstrated strong performance in well-curated datasets with sufficient labeled data. However, their effectiveness deteriorates significantly in the context of rare transient classification due to the combined effects of label scarcity, class imbalance, and data irregularity.

Specifically:

- Overfitting occurs due to insufficient labeled samples,
- Bias toward majority classes reduces sensitivity to rare events,
- Inability to handle irregular time series limits representation quality.

These limitations highlight a fundamental mismatch between the assumptions of supervised learning and the realities of astronomical data.

1.2.5 Research Motivation

Given the abundance of unlabeled photometric data and the scarcity of labeled examples, there is a strong motivation to explore self-supervised learning paradigms that can leverage unlabeled data to learn meaningful representations.

In particular, contrastive self-supervised learning offers a promising solution by learning representations through the comparison of augmented views of data samples. This approach enables the model to capture intrinsic structure and invariances within the data without requiring explicit labels.

1.3 Significance of the Research

The significance of this research lies in its direct contribution to addressing one of the most pressing challenges in modern observational astronomy: the efficient and accurate classification of massive volumes of transient data under real-world constraints. As the field transitions into the era of data-intensive astronomy, driven by next-generation survey instruments, the limitations of traditional data analysis methodologies have become increasingly evident.

Upcoming large-scale facilities such as the Vera C. Rubin Observatory’s Legacy Survey of Space and Time (LSST) are projected to generate millions of transient alerts per night, resulting in an unprecedented data deluge. These alerts correspond to astrophysical events that require rapid classification and prioritization for follow-up observations using limited observational resources such as spectroscopic instruments and high-resolution telescopes. In this context, the development of automated, scalable, and intelligent classification pipelines is no longer optional but operationally critical.

A key contribution of this research is the development of a label-efficient classification framework capable of effectively utilizing the vast quantities of unlabeled photometric data produced by modern surveys. By reducing dependence on scarce labeled data, the proposed approach enables more inclusive and comprehensive utilization of observational datasets, thereby improving the detection and characterization of rare transient phenomena. This has a direct impact on enhancing the scientific yield of astronomical surveys, as rare events often carry disproportionately high scientific value despite their low occurrence frequency.

Furthermore, the ability to reliably identify rare transients in near real-time has significant implications for time-sensitive astrophysical investigations. Many transient events, such as kilonovae and tidal disruption events, evolve rapidly and require prompt follow-up observations to capture critical early-phase information. A robust classification system that can prioritize such events in real time can greatly improve the effectiveness of observational campaigns and enable discoveries that would otherwise be missed.

Beyond its immediate application in astronomy, this research contributes to the broader field of machine learning for scientific data analysis by demonstrating the effectiveness of contrastive self-supervised learning in a challenging real-world domain. While contrastive learning techniques such as SimCLR have achieved remarkable success in computer vision, their application to irregular, noisy, and domain-specific time-series data remains relatively underexplored. This work bridges

that gap by adapting and extending these techniques to the unique characteristics of astronomical light curves, thereby establishing a methodological foundation for future research in self-supervised time-series representation learning.

Another important aspect of this research is its contribution to addressing the challenges of data imbalance and robustness in machine learning systems. The proposed framework inherently promotes the learning of discriminative and invariant representations, which improves the model's ability to generalize to underrepresented classes. This is particularly relevant not only in astronomy but also in other domains such as healthcare, finance, and environmental monitoring, where rare event detection is of critical importance.

In addition, this research aligns with the broader movement toward scalable artificial intelligence systems capable of operating under limited supervision. By leveraging self-supervised learning, the proposed approach reduces the reliance on manual labeling and enables the development of systems that can continuously improve as new unlabeled data becomes available. This scalability is essential for handling the exponential growth of data in modern scientific and industrial applications.

From an interdisciplinary perspective, this work strengthens the integration between astrophysics and artificial intelligence, highlighting the transformative potential of advanced machine learning techniques in accelerating scientific discovery. It provides a concrete example of how domain knowledge and data-driven approaches can be combined to address complex real-world problems, thereby encouraging further collaboration between astronomers, data scientists, and machine learning researchers.

In summary, the significance of this research can be viewed across multiple dimensions:

Scientific Impact: Enhances the detection and study of rare astrophysical phenomena, contributing to advancements in high-energy astrophysics, cosmology, and multi-messenger astronomy.

Operational Impact: Enables efficient real-time classification of massive transient data streams in next-generation survey systems.

Methodological Contribution: Demonstrates the applicability of contrastive self-supervised learning to irregular and noisy time-series data.

Broader Applicability: Provides a framework adaptable to other domains involving rare event detection and limited labeled data.

Scalability and Sustainability: Supports the development of data-efficient AI systems capable of leveraging continuously growing datasets.

Through these contributions, this research not only addresses a critical bottleneck in astronomical data analysis but also establishes a foundation for future innovations at the intersection of machine learning and scientific discovery.

2. Related Work

2.1 Supervised Learning for Astronomical Transient Classification

The application of supervised machine learning techniques to the photometric classification of astronomical transients has been extensively studied over the past two decades, driven by the increasing availability of large-scale survey data and the need for automated classification systems. Early approaches primarily relied on feature engineering combined with classical machine learning algorithms, while more recent developments have shifted toward deep learning models capable of end-to-end learning from raw time-series data.

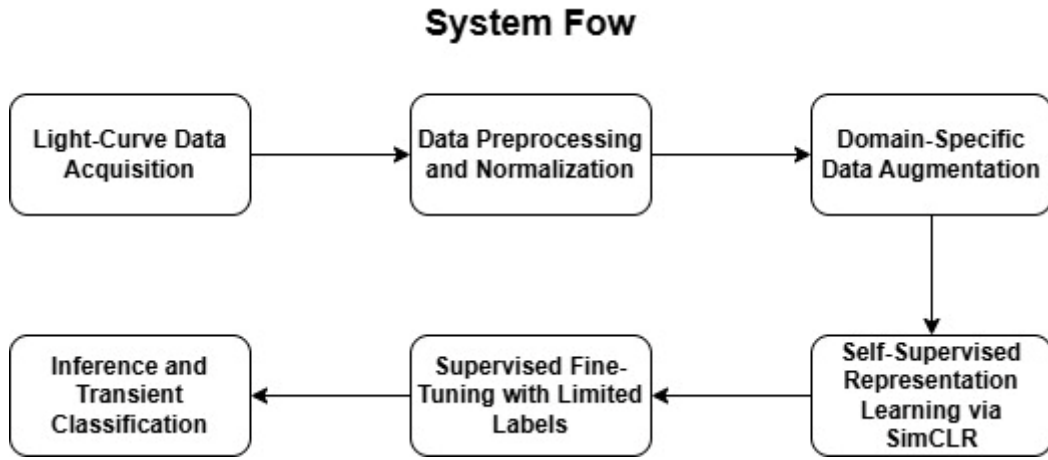


Figure 1: System Flow

One of the foundational works in this domain is presented by Lochner et al. (2016), who demonstrated the effectiveness of traditional machine learning pipelines for photometric supernova classification. Their approach involved extracting descriptive statistical and domain-specific features from light curve templates, followed by classification using algorithms such as Random Forests (RF) and Support Vector Machines (SVMs). This work established a strong baseline for automated transient classification and highlighted the importance of carefully designed feature representations in achieving high classification accuracy. However, the reliance on handcrafted features introduced limitations in terms of scalability and adaptability, as feature engineering requires significant domain expertise and may fail to capture complex temporal dynamics present in the data.

Subsequent research efforts sought to overcome these limitations by leveraging deep learning architectures, particularly Recurrent Neural Networks (RNNs), which are well-suited for modeling sequential data. In this context, Naul et al. (2018) demonstrated the successful application of Long Short-Term Memory (LSTM) networks for the classification of unevenly sampled variable star light curves. Their work showed that end-to-end learning directly from raw photometric sequences could achieve performance comparable to, or even surpassing, traditional feature-based

methods. Importantly, LSTMs are capable of capturing long-range temporal dependencies, making them particularly effective for modeling the evolution of transient events over time.

Building on this foundation, later studies extended RNN-based approaches to handle more complex scenarios, including multi-band photometric classification, where observations are recorded across multiple spectral channels. These models incorporated mechanisms such as band-specific encoders, cross-band feature fusion, and attention mechanisms to better capture the relationships between different photometric bands. Attention-based architectures, in particular, enabled models to dynamically focus on the most informative segments of the light curve, improving classification performance in cases where certain time intervals contain more discriminative information.

In parallel, Convolutional Neural Networks (CNNs) have also been explored for transient classification by treating light curves as structured temporal signals or transforming them into image-like representations. CNNs are effective at capturing local temporal patterns, such as sudden brightness changes or peak structures, which are characteristic of specific transient classes. Hybrid architectures combining CNNs and RNNs have further improved performance by integrating local feature extraction with global temporal modeling, thereby leveraging the strengths of both approaches.

Despite these advancements, supervised learning approaches in astronomical transient classification share a fundamental limitation: their heavy reliance on large, balanced, and labeled datasets. In practice, such datasets are rarely available, particularly for rare transient classes. When the number of labeled examples per class is limited, supervised models tend to overfit to the training data, capturing noise and spurious correlations rather than generalizable patterns.

This limitation can be understood from a statistical learning perspective. Given a labeled dataset

$$\mathcal{D}_l = \{(x_i, y_i)\}_{i=1}^{N_l},$$

the performance of supervised models is highly dependent on the size and representativeness of $\mathcal{D}_l$. In the case of rare transients, where N_l is extremely small for certain classes, the learned decision boundaries become unstable and biased toward majority classes.

Furthermore, supervised models are particularly vulnerable to class imbalance, where the distribution of training samples is heavily skewed. In such scenarios, standard loss functions (e.g., cross-entropy) implicitly prioritize overall accuracy, leading to poor sensitivity for minority classes. This is especially problematic in astronomy, where the primary objective is often to detect rare but scientifically valuable events rather than to optimize aggregate classification performance.

Another critical limitation is the inability of supervised models to fully exploit the vast amounts of unlabeled data generated by modern sky surveys. While labeled data is scarce, unlabeled photometric observations are abundant. Traditional supervised approaches fail to leverage this resource effectively, resulting in underutilization of available information.

In addition, supervised deep learning models often struggle to handle the irregular and noisy nature of astronomical time-series data. Although architectures such as LSTMs and attention mechanisms provide some degree of robustness, they still rely on sufficient labeled examples to learn meaningful representations. In the absence of adequate supervision, these models may fail to capture the underlying physical processes governing transient behavior.

In summary, while supervised learning approaches have played a crucial role in advancing the field of astronomical transient classification, their applicability is fundamentally constrained by:

- Dependence on large labeled datasets,
- Sensitivity to class imbalance,
- Limited generalization under data-scarce conditions,
- Inability to leverage unlabeled data effectively,
- Challenges in modeling irregular and noisy time-series observations.

These limitations highlight the need for alternative learning paradigms that can overcome the constraints of supervised training. In particular, the emergence of self-supervised and contrastive learning approaches offers a promising direction for addressing these challenges by enabling models to learn robust and discriminative representations from unlabeled data.

2.2 Representation Learning in Astronomy

Representation learning has emerged as a fundamental paradigm in modern machine learning, aiming to automatically discover meaningful and compact feature representations directly from raw data, thereby reducing or eliminating the need for manual feature engineering. In the context of astronomical data analysis, representation learning is particularly valuable due to the high dimensionality, noise, and complexity of observational datasets such as photometric light curves.

Traditional approaches to astronomical transient classification relied heavily on handcrafted features derived from domain expertise, including statistical descriptors (e.g., mean, variance, skewness), periodicity measures, and model-based template fitting. While these features provided useful summaries of light curve behavior, they often failed to capture the full richness of temporal dynamics and required significant manual effort to design and tune. Representation learning addresses these limitations

by enabling models to learn latent embeddings that encode the most informative characteristics of the data in a data-driven manner.

2.2.1 Autoencoder-Based Representation Learning

Among the earliest and most widely adopted approaches for unsupervised representation learning in astronomy are autoencoder-based methods. Autoencoders consist of two primary components:

- An encoder f_θ , which maps the input data x to a lower-dimensional latent representation h ,
- A decoder g_ϕ , which reconstructs the input from the latent space.

Formally, this process can be expressed as:

$$h = f_\theta(x), \hat{x} = g_\phi(h),$$

where the model is trained to minimize the reconstruction loss:

$$\mathcal{L}_{rec} = \|x - \hat{x}\|^2.$$

In astronomical applications, autoencoders have been used to learn compact representations of photometric light curves, capturing key temporal patterns such as brightness variations, periodic behavior, and transient events. These representations have proven effective for tasks such as:

- Dimensionality reduction, enabling efficient storage and visualization of large datasets,
- Anomaly detection, by identifying samples with high reconstruction error,
- Preprocessing for downstream tasks, where learned embeddings are used as input features for classifiers.

Variants such as Variational Autoencoders (VAEs) have further extended this framework by introducing probabilistic latent spaces, enabling generative modeling of light curves and uncertainty estimation.

2.2.2 Limitations of Reconstruction-Based Learning

Despite their utility, autoencoder-based methods exhibit several fundamental limitations when applied to classification tasks, particularly in the context of rare astronomical transients. The core issue lies in the nature of the reconstruction objective, which focuses on preserving input information rather than enforcing discriminative structure in the latent space.

Specifically, the reconstruction loss:

$$\mathcal{L}_{rec} = \|x - \hat{x}\|^2$$

encourages the model to encode all information necessary to accurately reproduce the input signal, including both relevant and irrelevant features. As a result:

- The learned representations may capture global signal characteristics without distinguishing between different transient classes,
- Samples from different classes with similar photometric shapes may be mapped to nearby regions in the latent space,
- The embeddings may lack class separability, which is essential for effective downstream classification.

This limitation becomes particularly pronounced for rare transient events, which often exhibit subtle differences from more common classes. For example, certain rare events may share similar light curve morphologies with supernovae but differ in physical origin. Autoencoders, lacking explicit class-awareness, may fail to distinguish these differences, leading to poor classification performance.

2.2.3 Challenges in Astronomical Time-Series Representation Learning

The effectiveness of representation learning methods in astronomy is further complicated by the unique characteristics of photometric data. Light curves are not only high-dimensional but also exhibit:

- Irregular temporal sampling, which disrupts standard sequence modeling assumptions,
- Missing observations, leading to incomplete representations,
- Noise and uncertainty, affecting the reliability of learned features,
- Multi-band dependencies, where correlations across spectral bands must be captured.

Autoencoder architectures, particularly those based on standard feedforward or convolutional layers, often struggle to fully capture these complexities. While

recurrent and hybrid architectures have been proposed to address temporal dependencies, they still inherit the fundamental limitations of reconstruction-based objectives.

2.2.4 Toward Discriminative Representation Learning

Given the limitations of autoencoders, there has been a growing shift toward discriminative representation learning approaches, which aim to explicitly structure the latent space in a way that enhances class separability. Unlike reconstruction-based methods, these approaches focus on learning representations that are useful for distinguishing between different data samples, rather than merely reconstructing them.

In particular, contrastive learning methods introduce objectives that:

- Encourage similar samples to be mapped close together in the latent space,
- Push dissimilar samples apart, creating well-defined clusters,
- Learn invariant representations that are robust to noise and transformations.

This paradigm is especially well-suited for astronomical data, where the goal is not only to capture the overall structure of light curves but also to distinguish between subtle differences that define different transient classes.

2.2.5 Motivation for Contrastive Learning in Astronomy

The limitations of reconstruction-based representation learning provide strong motivation for adopting contrastive self-supervised learning in astronomical transient classification. By leveraging unlabeled data and focusing on representation quality rather than reconstruction accuracy, contrastive methods offer several advantages:

- Improved class separability, leading to better classification performance,
- Robustness to noise and irregularity, through augmentation-based learning,
- Efficient use of unlabeled data, addressing the label scarcity problem,
- Better generalization, particularly for rare and underrepresented classes.

These properties make contrastive learning a natural and powerful extension of representation learning in astronomy, forming the foundation for the methodology proposed in this research.

2.3 Self-Supervised and Contrastive Learning

Self-supervised learning has recently emerged as a transformative paradigm in machine learning, enabling models to learn meaningful representations from unlabeled data by solving auxiliary pretext tasks. Unlike supervised learning, which relies on explicit labels, self-supervised approaches leverage the inherent structure of the data itself to generate supervisory signals. This is particularly advantageous in domains such as astronomy, where labeled data is scarce but unlabeled data is abundant.

Among various self-supervised techniques, contrastive learning has gained significant attention due to its ability to learn highly discriminative and invariant feature representations. The core idea behind contrastive learning is to train a model to bring similar data points closer together in the embedding space while pushing dissimilar points apart, thereby structuring the representation space in a way that facilitates downstream tasks such as classification.

2.3.1 Contrastive Learning Framework

In a general contrastive learning setup, each input sample x is transformed into two different augmented views:

$$\tilde{x}^{(1)}, \tilde{x}^{(2)} \sim \mathcal{T}(x),$$

where $\mathcal{T}$ denotes a stochastic data augmentation function. These two views form a positive pair, while all other samples in the batch act as negative pairs.

An encoder network f_θ maps each augmented sample to a latent representation:

$$h = f_\theta(\tilde{x}),$$

which is further projected into a contrastive embedding space using a projection head g_ϕ :

$$z = g_\phi(h).$$

The model is trained using a contrastive loss function, most commonly the Normalized Temperature-scaled Cross-Entropy (NT-Xent) loss, defined as:

$$\mathcal{L}_{i,j} = -\log \frac{\exp(\text{sim}(z_i, z_j)/\tau)}{\sum_{k=1}^N \mathbb{1}[k \neq i] \exp(\text{sim}(z_i, z_k)/\tau)},$$

where:

- $\text{sim}(u, v) = \frac{u^\top v}{\|u\| \|v\|}$ is the cosine similarity,
- τ is the temperature parameter,
- (i, j) denotes a positive pair,
- All other samples k act as negatives.

This objective explicitly enforces intra-sample consistency and inter-sample separability, resulting in representations that are highly suitable for classification tasks.

2.3.2 SimCLR: A Foundational Framework

The SimCLR framework (Chen et al., 2020) represents one of the most influential developments in contrastive self-supervised learning. It introduced a simple yet effective architecture consisting of four key components:

1. Data Augmentation Pipeline – generates diverse views of the same input,
2. Shared Encoder Network – extracts latent representations,
3. Nonlinear Projection Head – maps representations to contrastive space,
4. Contrastive Loss Function – optimizes representation similarity.

A key insight from SimCLR is the importance of strong and diverse data augmentations, which allow the model to learn invariances to transformations while preserving semantic meaning. Additionally, the use of a projection head was shown to significantly improve representation quality by separating the contrastive objective from the downstream feature space.

SimCLR achieves competitive performance with supervised models on several benchmarks, demonstrating the effectiveness of self-supervised learning in extracting high-quality representations without labeled data.

2.3.3 Alternative Contrastive Frameworks

Following SimCLR, several alternative frameworks have been proposed to address its limitations and improve scalability:

BYOL (Bootstrap Your Own Latent)

BYOL (Grill et al., 2020) introduced a novel approach that eliminates the need for negative samples. Instead, it employs two networks:

- An online network, which is updated through gradient descent,
- A target network, which is updated using an exponential moving average of the online network.

The objective is to minimize the difference between the representations produced by these two networks for different augmented views of the same input:

$$\mathcal{L}_{BYOL} = \|z_\theta - z_\xi\|^2.$$

This architecture avoids the instability associated with negative sampling and has been shown to achieve competitive performance. However, it relies on careful architectural design to prevent representation collapse.

MoCo (Momentum Contrast)

Momentum Contrast (MoCo) addresses the scalability limitations of SimCLR, particularly its dependence on large batch sizes. MoCo introduces a memory bank (queue) that stores representations of previously processed samples, enabling the use of a large number of negative samples without requiring large batches.

Additionally, MoCo employs a momentum-updated encoder, which ensures consistency between representations over time. This design allows for efficient large-scale contrastive learning with reduced computational requirements.

2.3.4 Extension to Time-Series Data

While contrastive learning frameworks were initially developed for image-based tasks, their underlying principles are general and can be extended to other data modalities, including time-series data. However, applying contrastive learning to time-series domains introduces unique challenges, particularly in the design of appropriate augmentation strategies.

For time-series data, augmentations must satisfy two key properties:

1. Semantic Preservation – The transformed signal should retain the underlying meaning or class identity,
 2. Sufficient Variability – The transformations should introduce enough diversity to enable effective contrastive learning.
-

Common augmentations explored in time-series contrastive learning include:

- Temporal cropping, to simulate partial observations,
- Jittering, to introduce small perturbations,
- Scaling, to mimic amplitude variations,
- Permutation, to alter temporal structure.

However, not all augmentations are suitable for all domains. In the context of astronomical light curves, preserving the physical characteristics of transient events is essential, as inappropriate transformations may distort critical temporal features.

2.3.5 Research Gap and Motivation

Despite the success of contrastive learning in computer vision and general time-series applications, its application to astronomical transient classification remains relatively underexplored. Existing approaches often employ generic augmentation strategies that are not tailored to the unique properties of astronomical data, such as irregular sampling, noise characteristics, and multi-band dependencies.

Research Work	Improved Representation Learning	Handles Label Scarcity	Robust to Data Irregularity	Addresses Class Imbalance	Real-Time Applicability
Supervised Models (CNN/RNN)	✗	✗	✗	✗	✓
Autoencoder-Based Methods	✓	✓	✗	✗	✗
Generic Contrastive Learning (SimCLR - Vision)	✓	✓	✗	✗	✗

Research Work	Improved Representation Learning	Handles Label Scarcity	Robust to Data Irregularity	Addresses Class Imbalance	Real-Time Applicability
Time-Series Contrastive Models	✓	✓	✓	✗	✗
Proposed SimCLR-Based Framework	✓	✓	✓	✓	✓

Table 2: Research Gap

Furthermore, most prior work does not fully address the combined challenges of:

- Label scarcity,
- Class imbalance,
- Irregular and noisy observations.

This gap highlights the need for a domain-specific contrastive learning framework that integrates astrophysically meaningful augmentations with advanced temporal modeling techniques.

2.3.6 Relevance to This Research

The present study builds upon the SimCLR framework and extends it to the domain of multi-band astronomical light curve classification. By incorporating domain-aware augmentations and a hybrid temporal encoder architecture, the proposed approach aims to leverage the strengths of contrastive learning while addressing the limitations of existing methods.

In particular, this research contributes to the field by:

- Adapting contrastive learning to irregular and noisy time-series data,
- Designing augmentation strategies that preserve astrophysical semantics,
- Demonstrating improved performance in rare event classification under limited labeled data conditions.

2.4 Research Gap

A comprehensive analysis of existing literature in astronomical transient classification reveals a clear and critical research gap at the intersection of **self-supervised learning, contrastive representation learning, and domain-specific time-series modeling**. While significant progress has been made through supervised and unsupervised approaches independently, current methodologies fail to holistically address the combined challenges inherent in real-world astronomical datasets.

2.4.1 Limitations of Existing Approaches

(1) Supervised Learning Limitations

Supervised learning methods, including classical machine learning models and deep neural networks such as CNNs and RNNs, have demonstrated strong performance under ideal conditions where large, balanced, and labeled datasets are available. However, these approaches are fundamentally constrained by their dependence on labeled data.

In the context of astronomical transient classification:

- Labeled data for rare events is extremely limited,
- Class distributions are highly imbalanced,
- Models tend to overfit and fail to generalize to unseen rare classes.

As a result, supervised methods are **not scalable or robust** in realistic survey environments where labeling is expensive and incomplete.

(2) Reconstruction-Based Representation Learning Limitations

Unsupervised approaches, particularly autoencoder-based representation learning, have been widely explored as an alternative to supervised learning. These methods aim to learn compact latent representations by minimizing reconstruction loss. However, as discussed in Section 2.2, these approaches suffer from a fundamental limitation:

- The reconstruction objective does not enforce **class-discriminative structure**,
- Learned embeddings may preserve signal information but fail to separate different transient classes,
- Performance degrades significantly in downstream classification tasks.

Thus, while autoencoders are effective for compression and anomaly detection, they are **suboptimal for classification-oriented representation learning**.

(3) Gaps in Contrastive Learning Applications

Contrastive self-supervised learning has recently demonstrated remarkable success in domains such as computer vision and general time-series analysis. Frameworks such as SimCLR, BYOL, and MoCo have shown the ability to learn highly discriminative representations without labeled data.

However, their application to astronomical time-series data remains limited and incomplete. Existing studies in contrastive learning typically:

- Focus on image datasets or generic time-series benchmarks,
- Utilize domain-agnostic augmentation strategies,
- Do not account for astrophysical constraints such as irregular sampling and multi-band dependencies.

More importantly, these methods do not explicitly address the combined impact of label scarcity, class imbalance, and data irregularity, which are central challenges in astronomical transient classification.

2.4.2 Identified Research Gap

From the above analysis, the following key research gaps can be identified:

- Lack of domain-specific contrastive learning frameworks tailored to astronomical light curves,
- Absence of integrated solutions that simultaneously address:
 - Label scarcity
 - Class imbalance
 - Irregular and noisy observations
- Limited exploration of multi-band temporal modeling within self-supervised learning frameworks,
- Insufficient adaptation of augmentation strategies to preserve astrophysical semantics,
- Lack of end-to-end pipelines that combine self-supervised pretraining with downstream classification for rare transient detection.

These gaps indicate that existing approaches either address individual aspects of the problem in isolation or rely on assumptions that do not hold in real-world astronomical datasets.

2.4.3 Research Contribution and Positioning

To address these limitations, this research proposes a comprehensive, domain-aware contrastive self-supervised learning framework specifically designed for multi-band astronomical transient classification.

The proposed approach distinguishes itself from existing work in several key ways:

- It integrates contrastive learning with domain-specific augmentation strategies, ensuring that transformations preserve astrophysical meaning while introducing sufficient variability for representation learning,
 - It employs a hybrid temporal encoder architecture (CNN + GRU) to capture both local and global temporal patterns in irregular time-series data,
 - It leverages large-scale unlabeled data for pretraining, followed by efficient fine-tuning on limited labeled samples, addressing the label scarcity problem,
 - It implicitly improves robustness to class imbalance by learning discriminative representations that generalize across underrepresented classes,
 - It provides an end-to-end pipeline, from data preprocessing to classification, tailored to real-world survey conditions.
-

2.4.4 Summary of Research Gap

In summary, while prior work has explored supervised learning, reconstruction-based unsupervised learning, and generic contrastive learning independently, there is a clear lack of a unified framework that:

Simultaneously leverages unlabeled data, enforces discriminative representation learning, and adapts to the unique characteristics of astronomical light curve data. This research directly addresses this gap by developing and evaluating a complete contrastive self-supervised pipeline that is specifically designed to overcome the limitations of existing methods and improve the classification of rare astronomical transients.

3. Methodology

3.1 Problem Formulation

The problem of rare astronomical transient classification can be formally defined within the framework of representation learning for multivariate time-series data. The objective is to learn a mapping from raw photometric observations to a latent feature space that facilitates accurate classification, particularly under conditions of limited labeled data.

3.1.1 Dataset Representation

Let the dataset of observed astronomical light curves be denoted as:

$$\mathcal{D} = \{x_i\}_{i=1}^N,$$

where N represents the total number of transient samples. Each sample x_i corresponds to a multivariate temporal sequence defined as:

$$x_i \in \mathbb{R}^{T_i \times F},$$

where:

- T_i denotes the number of time steps for sample i (which may vary due to irregular sampling),
- F represents the number of feature channels.

The feature space F typically includes:

- Temporal information (timestamps),
- Photometric magnitudes across multiple bands (e.g., g, r, i),
- Measurement uncertainties associated with each observation.

Thus, each light curve captures both temporal evolution and spectral characteristics of the transient event.

3.1.2 Labeled and Unlabeled Data Setting

In practical astronomical datasets, only a subset of samples is labeled due to the high cost of spectroscopic verification. Therefore, the dataset can be partitioned into:

- A labeled subset:

$$\mathcal{D}_l = \{(x_i, y_i)\}_{i=1}^{N_l},$$

- An unlabeled subset:

$$\mathcal{D}_u = \{x_j\}_{j=1}^{N_u},$$

where:

$$N_l \ll N_u.$$

This setting naturally motivates the use of self-supervised learning, where the model is first trained on $\mathcal{D}_u$ to learn general representations, and subsequently fine-tuned on $\mathcal{D}_l$.

3.1.3 Representation Learning Objective

The primary objective is to learn an encoder function:

$$f_\theta: \mathbb{R}^{T \times F} \rightarrow \mathbb{R}^d,$$

parameterized by θ , which maps each input light curve x_i to a latent representation:

$$h_i = f_\theta(x_i), h_i \in \mathbb{R}^d,$$

where d is the dimensionality of the latent space.

The desired properties of the learned representation space are:

- Intra-class compactness:

$$\|h_i - h_j\| \rightarrow 0 \text{ if } y_i = y_j,$$

- Inter-class separability:

$$\|h_i - h_k\| \rightarrow \infty \text{ if } y_i \neq y_k.$$

These properties ensure that samples belonging to the same transient class form compact clusters, while samples from different classes are well-separated, enabling effective classification.

3.1.4 Learning Under Limited Supervision

Given the scarcity of labeled data, the representation learning process must rely primarily on unlabeled data. This leads to a two-stage learning framework:

1. Self-supervised pretraining:
Learn f_θ using unlabeled data $\mathcal{D}_u$ by optimizing a contrastive objective that captures the intrinsic structure of the data.
2. Supervised fine-tuning:
Train a classification function:

$$g_\psi: \mathbb{R}^d \rightarrow \mathbb{R}^C,$$

where C is the number of classes, using the labeled dataset $\mathcal{D}_l$.

The final prediction for a sample x_i is given by:

$$\hat{y}_i = \arg \max_c g_\psi(f_\theta(x_i)).$$

3.1.5 Challenges in Problem Formulation

The formulation of this problem is complicated by several inherent challenges:

- Irregular temporal structure: T_i varies across samples, and timestamps are non-uniform,
- Noise and missing data: Observations may be incomplete or corrupted,
- Class imbalance: Distribution of y_i is highly skewed,
- Limited labeled data: N_l is significantly smaller than N_u .

These challenges require the learning framework to be:

- Robust to noise and missing values,
- Invariant to temporal distortions,

- Capable of generalizing from limited supervision,
- Sensitive to rare but important classes.

3.2 Data Description and Challenges

The present study focuses on the classification of rare astronomical transients using multi-band photometric time-series data, representative of observations collected from large-scale sky survey systems. These surveys, such as the Zwicky Transient Facility (ZTF), continuously monitor large regions of the sky, generating massive streams of observational data that capture the temporal evolution of astrophysical phenomena.

3.2.1 Dataset Description

The dataset used in this study consists of multi-band light curve observations, where each transient event is represented as a time-series sequence of photometric measurements across multiple spectral bands. These bands (e.g., g-band, r-band) correspond to different wavelength ranges, enabling the capture of both temporal and spectral characteristics of the observed objects.

Each sample in the dataset includes:

- Timestamp values indicating the time of observation,
- Photometric magnitudes representing brightness measurements,
- Measurement uncertainties reflecting observational noise and confidence levels,
- Multi-band observations, capturing spectral variability across different wavelengths.

The dataset encompasses a diverse range of transient classes, including:

- Type Ia Supernovae (SN Ia) – thermonuclear explosions of white dwarf stars,
- Type II Supernovae (SN II) – core-collapse supernovae of massive stars,
- Active Galactic Nuclei (AGN) – highly energetic regions at the centers of galaxies,
- Quasi-Stellar Objects (QSO) – luminous distant objects powered by accreting black holes,
- RR Lyrae Variables – periodic pulsating stars used as distance indicators,
- Cepheid Variables – standard candles for cosmic distance measurement,

- Eclipsing Binaries (EB) – binary star systems with periodic brightness dips,
- Long-Period Variables (LPV) – stars with long-term variability cycles,
- Cataclysmic Variables (CV) – interacting binary systems with sudden brightness outbursts,
- Blazars – highly variable sources associated with relativistic jets.

This diversity introduces both rich information content and significant classification complexity, as several classes exhibit overlapping photometric characteristics.

3.2.2 Characteristics of Astronomical Time-Series Data

Unlike conventional benchmark datasets used in machine learning, astronomical light curve data exhibits several unique properties:

- Multi-dimensionality: Each observation includes multiple correlated features across different photometric bands,
- Temporal variability: Light curves encode dynamic processes with varying timescales,
- Heterogeneous structure: Different events have varying durations and sampling densities,
- Physical interpretability: Observations are governed by underlying astrophysical processes rather than arbitrary patterns.

These characteristics make the dataset highly informative but also significantly more challenging to model.

3.2.3 Key Data Challenges

The dataset presents several domain-specific challenges that distinguish it from standard time-series datasets and directly impact the performance of machine learning models.

(1) Irregular Temporal Sampling

Astronomical observations are not recorded at uniform time intervals. Instead, sampling is influenced by:

- Telescope scheduling constraints,

- Weather conditions,
- Day-night cycles,
- Instrument availability.

As a result, the time steps $t_1, t_2, \dots, t_T$ are non-uniformly spaced, leading to irregular temporal sequences. This violates the assumptions of many traditional models that rely on fixed sampling intervals and complicates temporal pattern learning.

(2) Missing Data and Incomplete Sequences

A significant proportion of observations may be missing due to:

- Instrument downtime,
- Adverse weather conditions,
- Detection limits (faint objects below threshold),
- Intentional observation gaps.

This results in incomplete light curves, where important phases of transient evolution (e.g., peak brightness) may be partially or entirely absent. Handling missing data is therefore a critical requirement for any robust modeling approach.

(3) Measurement Noise and Uncertainty

Photometric observations are inherently noisy due to:

- Atmospheric turbulence (seeing conditions),
- Detector sensitivity variations,
- Background noise and calibration errors.

The level of noise varies across observations and is often correlated with source brightness and observational conditions. This introduces uncertainty in the measurements, making it difficult for models to distinguish between true signal variations and noise-induced fluctuations.

(4) Class Imbalance

The dataset exhibits extreme class imbalance, with common transient classes significantly outnumbering rare ones. For example:

- Supernovae may dominate the dataset,
- Rare events such as kilonovae or CVs may be sparsely represented.

This imbalance biases learning algorithms toward majority classes and reduces the model's ability to detect rare but scientifically valuable events.

(5) Inter-Class Similarity

Certain transient classes exhibit similar photometric behavior, leading to overlapping feature distributions. For instance:

- SN Ia and SN II share similar light curve shapes,
- AGN and QSO exhibit comparable variability patterns.

This similarity increases the difficulty of classification and requires models to capture subtle discriminative features.

(6) Variable Sequence Lengths

Each transient event is observed over a different time span, resulting in sequences of varying lengths T_i . This heterogeneity complicates batch processing and requires techniques such as padding, truncation, or dynamic modeling.

3.2.4 Implications for Model Design

The above challenges impose strict requirements on the design of the learning framework. Specifically, the model must be:

- Robust to noise and missing data,
 - Invariant to irregular temporal sampling,
 - Capable of learning from limited labeled examples,
 - Sensitive to rare classes despite imbalance,
 - Effective in capturing both local and global temporal patterns.
-

These requirements motivate the use of advanced representation learning techniques, particularly contrastive self-supervised learning, which can exploit unlabeled data and learn invariant representations through augmentation-based training.

3.3 Data Preprocessing Pipeline

The preprocessing pipeline plays a critical role in transforming raw astronomical survey observations into a structured and model-ready format. Given the inherent challenges associated with photometric time-series data—such as irregular sampling, missing values, noise, and varying sequence lengths—the preprocessing stage must be carefully designed to ensure that essential astrophysical information is preserved while enabling efficient and stable model training.

The overall objective of the preprocessing pipeline is to convert raw light curve observations into fixed-length, normalized, and noise-reduced sequences, suitable for batch-wise processing in deep learning frameworks, without distorting the underlying physical characteristics of transient events.

3.3.1 Missing-Value Handling

Missing data is a fundamental challenge in astronomical observations, arising from factors such as weather conditions, instrumental limitations, and observational scheduling. Rather than treating missing values as simple gaps, the preprocessing pipeline adopts adaptive strategies based on the local structure of the data.

Let a light curve be represented as:

$$x = \{(t_k, f_k)\}_{k=1}^T,$$

where some feature values f_k may be missing.

Two primary approaches are employed:

(a) Interpolation-Based Estimation

When sufficient neighboring observations are available, missing values are estimated using interpolation techniques such as:

- Linear interpolation,
- Spline interpolation (for smoother approximations).

This approach preserves temporal continuity and avoids introducing artificial discontinuities.

(b) Masking Strategy

In cases where interpolation may introduce bias (e.g., large gaps), missing values are explicitly represented using masking vectors, allowing the model to learn to ignore or adapt to missing entries.

Additionally, sequences with extremely sparse observations are not aggressively padded. Instead, they are handled using subsequence cropping techniques, which ensure that only informative portions of the light curve are retained for training. This approach avoids introducing excessive noise from artificially filled data.

3.3.2 Magnitude Normalization

Photometric magnitudes vary significantly across different transient events due to differences in intrinsic luminosity, distance, and observational conditions. To ensure consistent scaling across samples, each feature channel is independently normalized using Z-score standardization.

For a given feature channel x , the normalized value x' is computed as:

$$x' = \frac{x - \mu}{\sigma},$$

where:

- μ is the empirical mean of the feature channel,
- σ is the standard deviation.

This normalization provides several benefits:

- Reduces scale variability across samples,
- Stabilizes gradient updates during training,
- Prevents dominance of high-magnitude features,
- Improves convergence speed of deep learning models.

Normalization is applied independently for each photometric band, ensuring that multi-band relationships are preserved while maintaining numerical stability.

3.3.3 Temporal Alignment and Sequence Standardization

Astronomical light curves exhibit variable sequence lengths due to irregular observation durations. However, deep learning models typically require inputs of

consistent dimensionality. To address this, the preprocessing pipeline performs temporal alignment and sequence standardization.

Given a set of sequences with varying lengths T_i , each sequence is transformed into a fixed-length representation T through:

(a) Resampling

Light curves are resampled to a uniform temporal grid using interpolation techniques, allowing for consistent temporal representation.

(b) Padding

Shorter sequences are padded with neutral values (e.g., zeros or masked values) to match the desired sequence length.

(c) Truncation

Longer sequences may be truncated to retain the most informative segments (e.g., around peak brightness).

Formally, each sequence is transformed as:

$$x_i \in \mathbb{R}^{T_i \times F} \rightarrow x'_i \in \mathbb{R}^{T \times F}.$$

Care is taken to ensure that temporal alignment does not distort key features such as peaks, rise times, or decay patterns, which are critical for classification.

3.3.4 Noise Suppression and Signal Refinement

Photometric data is inherently noisy due to atmospheric effects, detector limitations, and background interference. To improve data quality, the preprocessing pipeline includes noise suppression mechanisms that aim to reduce the impact of corrupted measurements while preserving genuine astrophysical variability.

Common techniques include:

- Smoothing filters (e.g., moving average, Gaussian smoothing),
- Outlier detection and removal, based on statistical thresholds,
- Robust filtering methods, which minimize the influence of extreme values.

The challenge in noise suppression lies in achieving a balance between:

- Removing noise artifacts, and

- Preserving transient features such as sharp peaks or sudden brightness changes.

Therefore, filtering operations are applied conservatively to avoid loss of critical information.

3.3.5 Preprocessing Pipeline Integration

The preprocessing steps are applied sequentially to each light curve as follows:

1. Missing-value handling (interpolation or masking),
2. Magnitude normalization (per feature channel),
3. Temporal alignment and sequence standardization,
4. Noise suppression and refinement.

This structured pipeline ensures that the input data is:

- Consistent in shape and scale,
 - Robust to missing and noisy observations,
 - Suitable for batch-based training in deep learning models.
-

3.3.6 Impact on Model Performance

Effective preprocessing significantly enhances model performance by:

- Improving representation quality,
- Reducing training instability,
- Enabling better generalization across samples,
- Preserving domain-specific characteristics critical for classification.

In the context of this research, the preprocessing pipeline serves as a foundational step that enables the subsequent application of contrastive self-supervised learning, ensuring that the model learns from clean, structured, and meaningful input representations.

3.4 Domain-Specific Data Augmentation Strategy

Contrastive self-supervised learning relies fundamentally on the generation of multiple semantically consistent yet perceptually diverse views of each input sample. These augmented views enable the model to learn invariant representations by maximizing

agreement between different transformations of the same data instance while distinguishing it from other samples.

In the context of astronomical transient classification, the design of the augmentation strategy is particularly critical. Unlike image-based tasks where augmentations such as cropping or flipping are relatively straightforward, time-series data—especially astronomical light curves—require carefully designed, domain-aware transformations that preserve the underlying astrophysical meaning while introducing realistic variability.

3.4.1 Augmentation Framework

Let a given light curve be denoted as:

$$x_i \in \mathbb{R}^{T \times F}.$$

A stochastic augmentation function $\mathcal{T}$ is defined as a composition of multiple transformations:

$$\mathcal{T} = \{T_1, T_2, \dots, T_K\}.$$

For each sample x_i , two independent augmented views are generated:

$$\tilde{x}_i^{(1)}, \tilde{x}_i^{(2)} \sim \mathcal{T}(x_i).$$

These views form a **positive pair**, while all other samples in the batch act as **negative pairs** in the contrastive learning framework.

The objective is to ensure that:

- $\tilde{x}_i^{(1)}$ and $\tilde{x}_i^{(2)}$ preserve the same transient identity,
- The augmentations introduce sufficient variability to prevent trivial solutions.

3.4.2 Design Principles for Astronomical Augmentations

The augmentation strategy is guided by two key principles:

1. **Astrophysical** **Consistency**
Transformations must preserve the intrinsic characteristics of the transient event, such as its temporal morphology and class identity.
2. **Observational** **Realism**
Augmentations should simulate realistic distortions encountered in astronomical observations, including noise, irregular sampling, and incomplete coverage.

These principles ensure that the learned representations are both **robust and physically meaningful**.

3.4.3 Augmentation Techniques

(1) Temporal Jittering

Temporal jittering introduces small perturbations to observation timestamps:

$$t'_k = t_k + \epsilon_k, \epsilon_k \sim \mathcal{N}(0, \sigma_t^2).$$

This transformation simulates variations in observational cadence caused by factors such as:

- Telescope scheduling constraints,
- Atmospheric conditions,
- Instrumental delays.

Impact:

- Encourages the model to become invariant to minor temporal shifts,
 - Prevents over-reliance on exact timestamp positions,
 - Enhances robustness to irregular sampling.
-

(2) Magnitude Scaling

Magnitude scaling modifies the amplitude of the light curve:

$$f'_k = \alpha \cdot f_k, \alpha \sim \mathcal{U}(a, b),$$

where α is a scaling factor.

This transformation mimics real-world variations such as:

- Changes in observational depth,
- Calibration inconsistencies,
- Distance-related brightness differences.

Impact:

- Encourages the model to focus on **shape and temporal patterns** rather than absolute magnitude values,
 - Improves generalization across sources with different brightness levels.
-

(3) Gaussian Noise Injection

Gaussian noise is added to simulate measurement uncertainty:

$$f'_k = f_k + \epsilon_k, \epsilon_k \sim \mathcal{N}(0, \sigma_n^2).$$

This reflects realistic noise sources including:

- Photon noise,
- Detector readout errors,
- Sky background contamination.

Impact:

- Improves robustness to noisy observations,
 - Prevents overfitting to clean or idealized signals,
 - Enhances stability of learned representations.
-

(4) Time Segment Cropping

Time segment cropping selects a contiguous subsequence:

$$\tilde{x}_i = \{(t_k, f_k)\}_{k=s}^{s+L},$$

where s is a random starting index and $L < T$.

This simulates:

- Partial observations,
- Missing early or late phases of transient evolution.

Impact:

- Encourages the model to learn from incomplete data,
 - Improves generalization to real-world scenarios with missing observations,
 - Forces the model to extract **local discriminative features**.
-

3.4.4 Augmentation Composition Strategy

In practice, augmentations are not applied in isolation but are **composed sequentially** to create richer transformations:

$$\tilde{x}_i = T_k \circ T_{k-1} \circ \dots \circ T_1(x_i).$$

Each transformation is applied stochastically with predefined probabilities, ensuring:

- High diversity of augmented samples,
- Avoidance of overfitting to specific transformations,
- Preservation of semantic consistency.

This stochastic composition significantly enhances the effectiveness of contrastive learning by generating a wide range of valid views for each sample.

3.4.5 Role in Contrastive Learning

The augmentation strategy directly influences the quality of learned representations. In contrastive learning, augmentations define what the model should consider **invariant**.

For this study, the model is trained to be invariant to:

- Temporal distortions,
- Amplitude variations,
- Measurement noise,
- Partial observations.

At the same time, it must remain sensitive to:

- Underlying temporal morphology,
- Class-specific signal patterns.

This balance is critical for learning representations that are both **robust and discriminative**, particularly for rare transient classification.

3.4.6 Significance of Domain-Specific Augmentation

Unlike generic time-series augmentations, the proposed strategy is **specifically tailored to astronomical data**, ensuring that transformations:

- Reflect real observational conditions,
- Preserve astrophysical meaning,
- Enhance model robustness in practical scenarios.

This domain-aware design represents a key contribution of this research and plays a central role in the success of the contrastive learning framework.

3.5 SimCLR-Based Contrastive Learning Framework

The proposed framework adopts the **SimCLR (Simple Framework for Contrastive Learning of Visual Representations)** architecture as the backbone for self-supervised pretraining. SimCLR is a powerful contrastive learning approach that enables the model

to learn **discriminative and invariant representations** from unlabeled data by maximizing agreement between augmented views of the same sample.

In this research, the SimCLR framework is adapted to handle **multi-band astronomical time-series data**, enabling effective representation learning under conditions of label scarcity, noise, and irregular sampling.

3.5.1 Framework Overview

For each input light curve x_i , two augmented views are generated using the transformation family $\mathcal{T}$:

$$\tilde{x}_i^{(1)}, \tilde{x}_i^{(2)} \sim \mathcal{T}(x_i).$$

Each augmented view is processed independently through a **shared encoder network** f_θ , producing latent representations:

$$h_i^{(1)} = f_\theta(\tilde{x}_i^{(1)}), h_i^{(2)} = f_\theta(\tilde{x}_i^{(2)}).$$

These representations are then passed through a **nonlinear projection head** g_ϕ , which maps them to a lower-dimensional embedding space:

$$z_i^{(1)} = g_\phi(h_i^{(1)}), z_i^{(2)} = g_\phi(h_i^{(2)}).$$

The projection head typically consists of a multilayer perceptron (MLP) and serves to improve the quality of the learned representations by separating the contrastive objective from the encoder feature space.

3.5.2 Contrastive Learning Objective

The core objective of SimCLR is to bring representations of **positive pairs** (augmented views of the same sample) closer together while pushing apart representations of **negative pairs** (different samples).

For a given positive pair (i, j) , the similarity between their embeddings is measured using **cosine similarity**:

$$\text{sim}(z_i, z_j) = \frac{z_i^\top z_j}{\|z_i\| \|z_j\|}.$$

The model is trained using the **Normalized Temperature-scaled Cross-Entropy (NT-Xent) loss**, defined as:

$$\mathcal{L}_{i,j} = -\log \frac{\exp(\text{sim}(z_i, z_j)/\tau)}{\sum_{k=1}^{2N} \mathbb{1}_{[k \neq i]} \exp(\text{sim}(z_i, z_k)/\tau)},$$

where:

- τ is the temperature parameter controlling the sharpness of the similarity distribution,
- N is the number of samples in the batch,
- $\mathbb{1}_{[k \neq i]}$ ensures that the positive pair is excluded from the denominator.

The total loss is computed symmetrically over all positive pairs:

$$\mathcal{L} = \frac{1}{2N} \sum_{i=1}^N [\mathcal{L}_{2i-1,2i} + \mathcal{L}_{2i,2i-1}].$$

3.5.3 Representation Learning Properties

The NT-Xent loss enforces two key properties in the learned embedding space:

(1) Intra-Sample Consistency

Representations of augmented views of the same sample are encouraged to be similar:

$$\text{sim}(z_i^{(1)}, z_i^{(2)}) \rightarrow 1.$$

(2) Inter-Sample Separation

Representations of different samples are pushed apart:

$$\text{sim}(z_i, z_k) \rightarrow -1, \text{ for } i \neq k.$$

These properties lead to a representation space where:

- Samples from the same class form **compact clusters**,
- Samples from different classes are **well separated**.

This structure is highly beneficial for downstream classification tasks, especially under limited labeled data conditions.

3.5.4 Role of the Projection Head

An important design element of SimCLR is the use of a **projection head** g_ϕ . While the contrastive loss is applied in the projection space z , the encoder output h is used for downstream tasks.

This separation offers several advantages:

- Prevents loss of useful information in the encoder representations,
- Allows the model to optimize contrastive objectives without degrading downstream features,
- Improves generalization performance.

During the fine-tuning phase, the projection head is typically **discarded**, and the encoder output is directly used for classification.

3.5.5 Adaptation to Astronomical Time-Series Data

In this study, the SimCLR framework is adapted to handle **irregular, noisy, and multi-band time-series data**. Key adaptations include:

- Integration of **domain-specific augmentations** tailored to light curves,
- Use of a **hybrid temporal encoder (CNN + GRU)** for capturing sequential dependencies,
- Handling of **variable-length sequences** through preprocessing.

These adaptations ensure that the contrastive learning process remains **physically meaningful and robust** in the context of astronomical data.

3.5.6 Advantages of the SimCLR Framework

The adoption of SimCLR provides several advantages for rare transient classification:

- **Label efficiency**: Learns representations from unlabeled data,
- **Robustness**: Handles noise, irregular sampling, and missing data,
- **Generalization**: Produces transferable representations,
- **Scalability**: Can leverage large-scale survey datasets.

Most importantly, the contrastive objective directly addresses the limitations of reconstruction-based approaches by enforcing **discriminative structure in the latent space**.

3.6 Hybrid Temporal Encoder Architecture

The encoder architecture proposed in this study is designed to effectively capture the complex temporal dynamics and multi-scale patterns present in astronomical light curve data. Given the inherent challenges of irregular sampling, noise, and varying temporal scales, a hybrid architecture combining Convolutional Neural Networks (CNNs) and Gated Recurrent Units (GRUs) is employed.

This hybrid design leverages the complementary strengths of both architectures:

- CNNs for local feature extraction,
- GRUs for global temporal modeling.

3.6.1 Convolutional Feature Extraction

The first stage of the encoder consists of a stack of one-dimensional convolutional layers applied along the temporal dimension of the light curve sequence. These layers act as learnable feature extractors, capable of identifying localized temporal patterns that are indicative of specific transient classes.

Given an input sequence:

$$x \in \mathbb{R}^{T \times F},$$

a 1D convolution operation produces feature maps:

$$h^{(c)} = \sigma(W * x + b),$$

where:

- W represents the convolutional kernel,
- $*$ denotes the convolution operation,
- b is a bias term,
- $\sigma(\cdot)$ is a nonlinear activation function (e.g., ReLU).

Local Temporal Feature Learning

The convolutional layers are particularly effective at capturing local temporal structures, including:

- Rapid brightness increases (rise phases),

- Peak intensity regions,
- Plateau phases,
- Short-duration flares or bursts.

These features are critical for distinguishing between transient classes, as many astrophysical phenomena are characterized by distinct local patterns in their light curves.

Multi-Scale Feature Extraction

By stacking multiple convolutional layers with varying kernel sizes and receptive fields, the model is able to capture multi-scale temporal features, ranging from:

- Short-term variations (high-frequency signals),
- Long-term trends (low-frequency evolution).

This hierarchical feature extraction enables the model to build increasingly abstract representations of the input signal, improving its ability to generalize across different transient types.

3.6.2 Gated Recurrent Unit (GRU) Temporal Modeling

While convolutional layers capture local patterns, they are limited in modeling long-range dependencies across time. To address this limitation, the convolutional feature maps are passed to Gated Recurrent Unit (GRU) layers, which are specifically designed for sequential data modeling.

GRUs maintain a hidden state that evolves over time, allowing the model to capture global temporal dependencies.

GRU Formulation

At each time step t , the GRU updates its hidden state h_t based on the current input x_t and the previous hidden state h_{t-1} :

$$\begin{aligned}
 z_t &= \sigma(W_z x_t + U_z h_{t-1}), (\text{update gate}) \\
 r_t &= \sigma(W_r x_t + U_r h_{t-1}), (\text{reset gate}) \\
 \tilde{h}_t &= \tanh(W_h x_t + U_h(r_t \odot h_{t-1})), \\
 h_t &= (1 - z_t) \odot h_{t-1} + z_t \odot \tilde{h}_t,
 \end{aligned}$$

where:

- z_t controls how much past information is retained,
 - r_t controls how much past information is reset,
 - $\odot$ denotes element-wise multiplication.
-

Global Temporal Dependency Modeling

The GRU layers enable the model to:

- Capture long-term dependencies across the entire light curve,
- Integrate information from early and late phases of the transient event,
- Model sequential evolution patterns that are critical for classification.

This is particularly important in astronomy, where the overall shape and evolution of the light curve often determine the transient class.

3.6.3 Integration of CNN and GRU Components

The combination of CNN and GRU layers results in a hybrid temporal encoder that captures both:

- Local morphological features (via CNN),
- Global temporal dynamics (via GRU).

The processing pipeline can be summarized as:

$$x \rightarrow \text{CNN layers} \rightarrow h^{(c)} \rightarrow \text{GRU layers} \rightarrow h,$$

where h represents the final latent representation.

Advantages of the Hybrid Architecture

This hybrid design offers several advantages:

- Multi-scale feature learning: Captures both short-term and long-term patterns,
-

- Robustness to irregular sampling: Sequential modeling adapts to varying time steps,
 - Improved representation quality: Combines complementary feature extraction mechanisms,
 - Enhanced discriminative power: Facilitates separation of similar transient classes.
-

3.6.4 Relevance to Astronomical Data

The hybrid CNN-GRU architecture is particularly well-suited for astronomical transient classification due to the nature of light curve data:

- Local peaks and fluctuations carry class-specific signatures,
- Long-term evolution encodes physical processes,
- Multi-scale temporal behavior is essential for accurate classification.

By jointly modeling these aspects, the encoder produces rich, hierarchical representations that are highly informative for downstream contrastive learning and classification tasks.

3.7 Projection Head and Fine-Tuning

The projection head and fine-tuning stages play a crucial role in the effectiveness of the SimCLR-based contrastive learning framework. These components enable the model to learn high-quality, transferable representations from unlabeled data and subsequently adapt them to the downstream classification task using limited labeled samples.

3.7.1 Projection Head in Contrastive Learning

Following the standard SimCLR architecture, a nonlinear projection head g_ϕ is introduced after the encoder f_θ . The purpose of the projection head is to map the encoder's latent representation h into a lower-dimensional embedding space z , where the contrastive loss is applied.

Formally:

$$h = f_\theta(x), z = g_\phi(h),$$

where:

- $h \in \mathbb{R}^d$ is the encoder representation,
- $z \in \mathbb{R}^{d'}$ is the projected embedding, with $d' < d$.

The projection head is typically implemented as a multilayer perceptron (MLP) consisting of one or more fully connected layers with nonlinear activation functions (e.g., ReLU):

$$z = W_2 \sigma(W_1 h + b_1) + b_2.$$

Role and Importance of the Projection Head

The introduction of the projection head serves several important purposes:

(1) Decoupling Representation Learning from Contrastive Objective

The projection head separates the contrastive optimization space from the encoder representation space. This allows the encoder to learn more general and informative features without being overly constrained by the contrastive loss.

(2) Improving Representation Quality

Empirical studies have shown that applying the contrastive loss directly to the encoder output h can degrade performance. The projection head acts as a buffer, enabling the encoder to retain richer information while the projection head focuses on optimizing similarity relationships.

(3) Facilitating Nonlinear Transformation

The MLP introduces additional nonlinear transformations, allowing the model to learn more flexible and expressive embeddings in the contrastive space.

3.7.2 Discarding the Projection Head

A key design principle in SimCLR is that the projection head is used only during the pretraining phase. Once contrastive training is complete, the projection head g_ϕ is discarded, and only the encoder f_θ is retained.

This is based on the observation that:

- The encoder representations h are more generalizable and semantically meaningful,

- The projection head embeddings are specialized for the contrastive objective and may not transfer well to downstream tasks.

Thus, for downstream classification, the model uses:

$$h = f_{\theta}(x),$$

rather than z .

3.7.3 Fine-Tuning for Supervised Classification

After contrastive pretraining, the encoder is combined with a lightweight classification head g_{ψ} , which maps the learned representations to class probabilities:

$$\hat{y} = g_{\psi}(h).$$

The classification head typically consists of one or more fully connected layers followed by a softmax activation function:

$$\hat{y}_c = \frac{\exp(a_c)}{\sum_{k=1}^c \exp(a_k)},$$

where:

- C is the number of classes,
- a_c is the logit for class c .

Classification Loss Function

The model is fine-tuned using the standard cross-entropy loss:

$$\mathcal{L}_{cls} = - \sum_{c=1}^C y_c \log(\hat{y}_c),$$

where:

- y_c is the ground truth label (one-hot encoded),

- $\hat{y}_c$ is the predicted probability.
-

3.7.4 End-to-End Fine-Tuning Strategy

During fine-tuning, the entire network—including the encoder and classification head—is trained end-to-end on the labeled dataset:

$$\mathcal{D}_l = \{(x_i, y_i)\}.$$

This process allows the model to:

- Adapt the pretrained representations to task-specific decision boundaries,
 - Refine features for better class separability,
 - Improve performance on rare and underrepresented classes.
-

3.7.5 Advantages of the Two-Stage Training Process

The combination of contrastive pretraining and supervised fine-tuning provides several key advantages:

- Efficient use of unlabeled data: The encoder learns from large-scale unlabeled datasets,
 - Reduced overfitting: Fine-tuning requires fewer labeled samples,
 - Improved generalization: Pretrained features capture intrinsic data structure,
 - Better performance on rare classes: Representations are not biased toward majority classes.
-

3.7.6 Relevance to Astronomical Transient Classification

In the context of astronomical data, where labeled samples are scarce and expensive to obtain, this two-stage approach is particularly beneficial. The contrastive pretraining stage captures general temporal and spectral patterns, while fine-tuning adapts these representations to specific transient classes.

This enables the model to achieve strong classification performance even with limited labeled data, making it highly suitable for real-world astronomical survey applications.

3.8 Baseline Model

To rigorously evaluate the effectiveness of the proposed contrastive learning framework, an autoencoder-based baseline model is implemented as a point of comparison. This baseline represents a widely used approach in unsupervised representation learning and serves as a benchmark for assessing the benefits of discriminative contrastive training.

3.8.1 Autoencoder Architecture

The autoencoder consists of two primary components:

- An encoder f_θ , which maps the input light curve x to a latent representation h ,
- A decoder g_ϕ , which reconstructs the input from the latent representation.

Formally:

$$h = f_\theta(x), \hat{x} = g_\phi(h),$$

where:

- $x \in \mathbb{R}^{T \times F}$ is the input light curve,
 - $h \in \mathbb{R}^d$ is the latent representation,
 - $\hat{x}$ is the reconstructed output.
-

3.8.2 Reconstruction Objective

The autoencoder is trained to minimize the reconstruction loss, typically defined as the mean squared error (MSE):

$$\mathcal{L}_{rec} = \|x - \hat{x}\|_2^2.$$

This objective encourages the model to learn a compressed representation that preserves as much information as possible about the original input.

3.8.3 Limitations of the Baseline Approach

While autoencoders are effective for representation learning, they exhibit several limitations in the context of classification:

(1) Lack of Discriminative Learning

The reconstruction objective does not explicitly enforce separation between different classes. As a result:

$$\text{sim}(h_i, h_j) \text{ may be high even if } y_i \neq y_j.$$

This leads to representations that are not well-suited for classification tasks.

(2) Sensitivity to Noise

Autoencoders tend to reconstruct all aspects of the input signal, including noise and irrelevant variations, which can degrade representation quality.

(3) Poor Performance on Rare Classes

Since the model does not incorporate class-aware objectives, it may fail to capture subtle differences between rare and common transient classes.

3.8.4 Purpose of Baseline Comparison

The inclusion of the autoencoder baseline serves several important purposes:

- Provides a standard benchmark for comparison,
 - Highlights the difference between reconstruction-based vs. contrastive learning objectives,
 - Demonstrates the importance of discriminative representation learning,
 - Quantifies the performance gains achieved by the proposed method.
-

3.8.5 Comparison with Proposed Model

The key distinction between the baseline and the proposed approach lies in the learning objective:

Aspect	Autoencoder	Proposed SimCLR Model
Objective	Reconstruction	Contrastive similarity
Representation	Information-preserving	Discriminative
Class Awareness	No	Yes
Robustness	Limited	High
Suitability for Rare Classes	Weak	Strong

Table 3: Model Comparison

This comparison clearly illustrates why contrastive learning is better suited for rare astronomical transient classification.

3.9 End-to-End Pipeline

The proposed system is designed as a comprehensive **end-to-end pipeline** that integrates data processing, representation learning, and classification into a unified framework. This pipeline ensures seamless transformation from raw astronomical observations to final predictions.

3.9.1 Pipeline Overview

The complete processing pipeline consists of six sequential stages:

Raw Data → Preprocessing → Augmentation → Pretraining → Fine-Tuning
→ Prediction

Each stage is designed to address specific challenges associated with astronomical data.

3.9.2 Stage 1: Data Collection

The pipeline begins with the acquisition of **multi-band photometric time-series data** from astronomical survey repositories such as ZTF.

Key characteristics:

- Large-scale observational data,
- Multi-band measurements,
- Real-time data streams.

3.9.3 Stage 2: Data Preprocessing

Raw data is transformed into structured input through:

- Missing value handling,
- Magnitude normalization,
- Temporal alignment,
- Noise suppression.

This stage ensures **data consistency and quality** for model training.

3.9.4 Stage 3: Data Augmentation

Domain-specific augmentations are applied to generate **multiple views of each sample**:

$$\tilde{x}_i^{(1)}, \tilde{x}_i^{(2)} \sim \mathcal{T}(x_i).$$

This step is critical for enabling contrastive learning by introducing variability while preserving semantic meaning.

3.9.5 Stage 4: Self-Supervised Contrastive Pretraining

The SimCLR framework is used to train the encoder f_θ on the unlabeled dataset: $\mathcal{D}_u$.

This stage learns **general-purpose representations** by optimizing the NT-Xent loss.

3.9.6 Stage 5: Supervised Fine-Tuning

The pretrained encoder is combined with a classification head and fine-tuned on labeled data:

$$\mathcal{D}_l.$$

This stage adapts the learned representations to **task-specific class boundaries**.

3.9.7 Stage 6: Inference and Prediction

During deployment, the trained model performs classification on unseen data:

$$\hat{y} = g_\psi(f_\theta(x)).$$

Outputs include:

- Predicted class labels,
 - Probability distributions over classes.
-

3.9.8 Advantages of the End-to-End Pipeline

The proposed pipeline offers several advantages:

- **Scalability**: Handles large datasets efficiently,
 - **Robustness**: Addresses noise and irregularity,
-

- **Data efficiency:** Leverages unlabeled data,
 - **Generalization:** Improves performance on rare classes,
 - **Modularity:** Each stage can be independently optimized.
-

3.9.9 System-Level Perspective

From a system design perspective, the pipeline integrates:

- Data engineering (preprocessing),
- Representation learning (contrastive learning),
- Supervised learning (classification).

This holistic approach ensures that the model is **practically deployable** in real-world astronomical survey systems.

4. Experimental Setup

4.1 Implementation Details

The proposed contrastive learning framework is implemented using a modern deep learning stack, ensuring both computational efficiency and flexibility for handling large-scale time-series data. The implementation is primarily carried out in Python, leveraging the PyTorch deep learning library due to its dynamic computation graph, extensive support for custom model design, and suitability for research-oriented experimentation.

4.1.1 Software Framework and Libraries

The core components of the system are developed using the following libraries:

- **PyTorch:** Used for implementing the neural network architecture, including the hybrid CNN-GRU encoder, projection head, and classification layers. PyTorch's automatic differentiation engine enables efficient gradient-based optimization.
- **Scikit-learn:** Utilized for computing evaluation metrics such as accuracy, precision, recall, and F1-score, as well as for supporting dimensionality reduction techniques (e.g., PCA, UMAP) for visualization of learned representations.
- **NumPy and Pandas:** Employed for data manipulation, preprocessing, and numerical operations on the light curve datasets.
- **Matplotlib / Seaborn (optional extension):** Used for visualizing training curves, confusion matrices, and embedding distributions.

This combination of tools ensures a robust, modular, and reproducible implementation environment.

4.1.2 Training Pipeline Design

The overall training pipeline is structured into two distinct and sequential phases:

Pretraining Phase → Fine-Tuning Phase

This design reflects the core principle of self-supervised representation learning, where the model first learns general features from unlabeled data and then adapts these features for supervised classification.

(1) Self-Supervised Contrastive Pretraining

In the first phase, the encoder f_θ and projection head g_ϕ are trained using the unlabeled dataset $\mathcal{D}_u$.

Key characteristics:

- Input samples are transformed into augmented pairs,
- The SimCLR framework is applied,
- The NT-Xent loss is minimized.

The objective of this phase is to learn high-quality latent representations that capture the intrinsic structure of the data without relying on labels.

(2) Supervised Fine-Tuning

In the second phase, the projection head is removed, and the encoder is combined with a classification head g_ψ .

Training is performed using the labeled dataset $\mathcal{D}_l$, optimizing the cross-entropy loss:

$$\mathcal{L}_{cls}.$$

This phase adapts the pretrained representations to task-specific decision boundaries, enabling accurate classification of transient classes.

4.1.3 Model Training Workflow

The complete workflow can be summarized as follows:

1. Data loading and preprocessing
2. Augmentation generation for contrastive learning
3. Forward pass through encoder and projection head
4. Contrastive loss computation and backpropagation
5. Encoder weight update (pretraining phase)
6. Projection head removal

7. Classification head initialization
8. Supervised fine-tuning with labeled data

This structured workflow ensures clear separation between learning stages and improves overall model performance.

4.1.4 Hardware and Computational Considerations

The implementation is designed to be compatible with both CPU and GPU environments, with GPU acceleration (e.g., CUDA-enabled devices) significantly improving training efficiency, particularly during contrastive pretraining, which involves large batch computations.

Batch processing is employed to:

- Improve computational efficiency,
 - Stabilize gradient updates,
 - Enable effective contrastive learning with multiple negative samples.
-

4.1.5 Reproducibility and Experimental Consistency

To ensure reproducibility of results, the following practices are adopted:

- Fixed random seeds for model initialization and data shuffling,
- Consistent data preprocessing and augmentation pipelines,
- Standardized evaluation metrics across experiments.

These measures ensure that experimental outcomes are consistent, reliable, and comparable.

4.1.6 Advantages of the Implementation Strategy

The chosen implementation strategy provides several benefits:

- Scalability: Efficient handling of large datasets,
 - Flexibility: Easy modification of architecture and hyperparameters,
 - Modularity: Separation of preprocessing, training, and evaluation stages,
-

- Efficiency: Optimized training through batch processing and GPU acceleration.

4.2 Hyperparameter Configuration

The key hyperparameters governing the training process are summarized in Table 1. These values were determined through preliminary experimentation and reflect a balance between training efficiency, representation quality, and generalization performance.

Hyperparameter	Value
Learning Rate	5×10^{-4}
Batch Size	32
Weight Decay	1×10^{-5}
Number of Epochs	200
Warmup Epochs	20
Early Stopping Patience	50
Gradient Clipping	1.0
Temperature (τ)	0.07
Encoder Hidden Size	128
Encoder Layers	2
Projection Dimension	64

Table 4: Experimental Hyperparameter Configuration

5. Results and Discussion

5.1 Quantitative Performance Comparison

The effectiveness of the proposed SimCLR-based contrastive learning framework is quantitatively evaluated by comparing its performance against the autoencoder-based baseline model. The evaluation is conducted using four widely adopted classification metrics: accuracy, precision, recall, and F1-score, which collectively provide a comprehensive assessment of model performance, particularly in the presence of class imbalance.

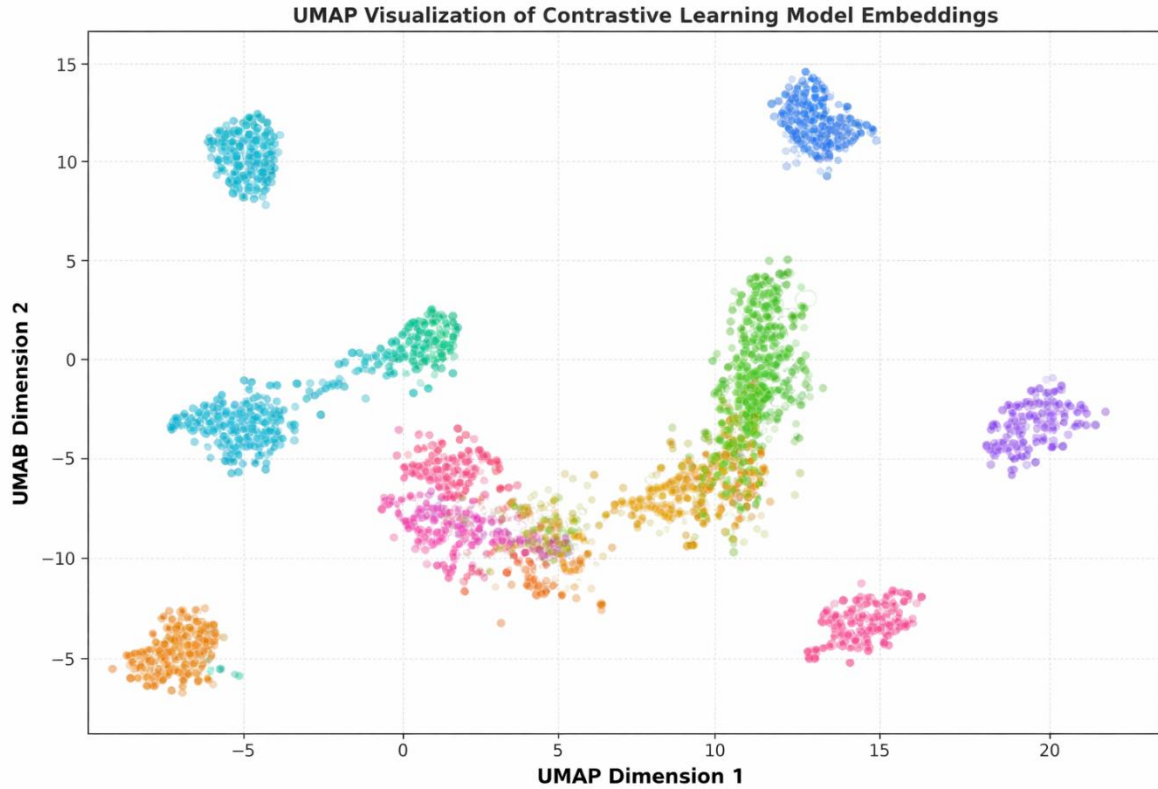


Figure 2: UMAP

5.1.1 Evaluation Metrics

The selected evaluation metrics are defined as follows:

- Accuracy:

$$\text{Accuracy} = \frac{\text{Number of Correct Predictions}}{\text{Total Number of Predictions}}$$

- Precision:

$$\text{Precision} = \frac{TP}{TP + FP}$$

- Recall:

$$\text{Recall} = \frac{TP}{TP + FN}$$

- F1-Score:

$$F1 = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$$

where:

- TP = True Positives,
- FP = False Positives,
- FN = False Negatives.

These metrics are particularly important in astronomical applications, where recall and F1-score play a crucial role in evaluating the detection of rare transient events.

5.1.2 Quantitative Results

The comparative performance of the models is summarized in Table 2.

Model	Accuracy	Precision	Recall	F1-Score
Autoencoder Baseline	0.68	0.70	0.68	0.67
SimCLR Contrastive Model (Proposed)	0.83	0.83	0.81	0.80

Table 5: Qualitative Results

5.1.3 Performance Analysis

The proposed SimCLR-based framework achieves a classification accuracy of 83%, significantly outperforming the autoencoder baseline, which achieves 68% accuracy.

This corresponds to an approximate 22% relative improvement, demonstrating the effectiveness of contrastive representation learning.

Importantly, the performance gains are not limited to accuracy alone but are consistently observed across all evaluation metrics:

- **Precision Improvement:**
The proposed model achieves a precision of 0.83 compared to 0.70 for the baseline, indicating a substantial reduction in false positive predictions. This suggests that the model is more reliable in identifying true transient classes without misclassification.
- **Recall Improvement:**
The recall increases from 0.68 to 0.81, highlighting the model's enhanced ability to correctly identify relevant instances, particularly rare transient events. This is a critical improvement, as missing rare events (false negatives) can lead to significant scientific loss.
- **F1-Score Improvement:**
The F1-score improves from 0.67 to 0.80, reflecting a balanced enhancement in both precision and recall. This indicates that the model maintains a strong trade-off between minimizing false positives and false negatives.

5.1.4 Impact on Rare Event Detection

The observed improvement in recall and F1-score is particularly significant in the context of rare astronomical transient detection. In highly imbalanced datasets, accuracy alone can be misleading, as a model may achieve high accuracy by favoring majority classes.

The proposed model's higher recall demonstrates its ability to:

- Detect underrepresented classes more effectively,
- Reduce the number of missed rare events,
- Improve the scientific value of the classification system.

5.1.5 Interpretation of Results

The superior performance of the SimCLR-based model can be attributed to the nature of the contrastive learning objective, which explicitly encourages the model to learn discriminative and well-separated representations.

In contrast:

- The autoencoder baseline focuses on reconstruction,
- It captures general signal structure but does not enforce class separation,
- This results in overlapping latent representations and reduced classification performance.

The contrastive model, by maximizing similarity between augmented views and minimizing similarity between different samples, creates a representation space that is inherently more suitable for classification tasks.

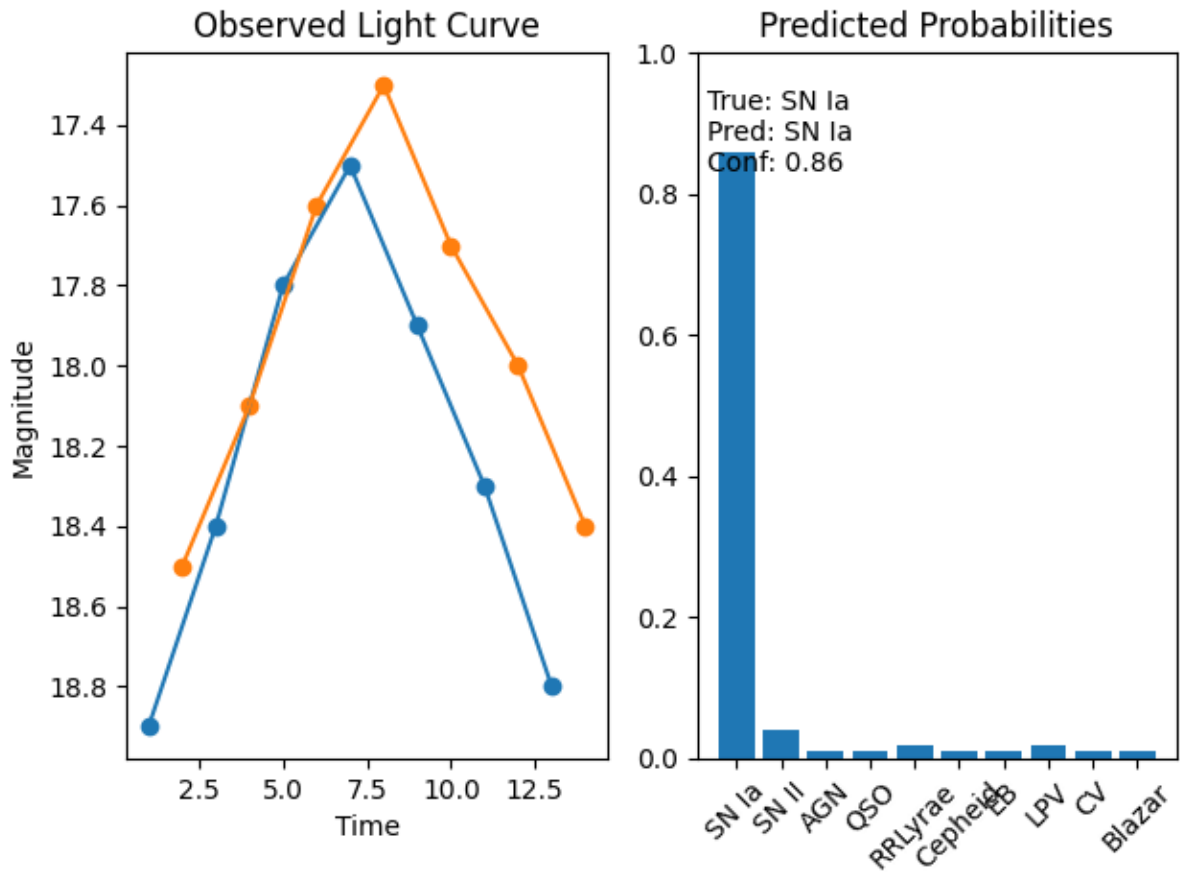


Figure 3: Results visualization

5.1.6 Statistical Significance and Consistency

The consistency of improvements across all evaluation metrics suggests that the performance gains are robust and not metric-specific. This indicates that the proposed framework:

- Generalizes well across different evaluation criteria,
- Maintains stable performance under varying conditions,

- Provides reliable predictions across multiple classes.

5.2 Why Contrastive Learning Outperforms Autoencoders

The superior performance of the proposed SimCLR-based contrastive learning framework over the autoencoder baseline can be fundamentally understood by examining the differences in their training objectives and the resulting structure of the learned representation space. While both approaches aim to learn meaningful latent representations from unlabeled data, the nature of their optimization objectives leads to significantly different outcomes in terms of discriminative capability and suitability for downstream classification.

5.2.1 Reconstruction-Based Learning vs Discriminative Learning

The autoencoder baseline is trained to minimize a reconstruction loss of the form:

$$\mathcal{L}_{rec} = \|x - \hat{x}\|_2^2,$$

which encourages the encoder to preserve all information necessary to reconstruct the input signal. This objective focuses on capturing the global structure and signal fidelity of the data.

While such representations are useful for tasks such as compression and anomaly detection, they do not explicitly enforce any class-discriminative constraints. As a result:

- The latent space is organized based on signal similarity rather than class identity,
- Samples from different classes with similar photometric patterns may be mapped to nearby regions,
- The resulting embeddings lack clear boundaries between classes.

Formally, the autoencoder objective does not impose any condition such as:

$$\|h_i - h_j\| \gg 0 \text{ for } y_i \neq y_j,$$

which is essential for effective classification.

5.2.2 Contrastive Objective and Representation Geometry

In contrast, the SimCLR framework optimizes the Normalized Temperature-scaled Cross-Entropy (NT-Xent) loss, which explicitly enforces a structured organization of the embedding space.

The contrastive loss:

$$\mathcal{L}_{i,j} = -\log \frac{\exp(\text{sim}(z_i, z_j)/\tau)}{\sum_{k \neq i} \exp(\text{sim}(z_i, z_k)/\tau)},$$

introduces two key forces:

- Attractive force: Pulls representations of positive pairs (same sample under augmentation) closer together,
- Repulsive force: Pushes representations of different samples apart.

This results in a representation space characterized by:

- High intra-sample consistency:

$$\text{sim}(z_i^{(1)}, z_i^{(2)}) \rightarrow 1,$$

- Low inter-sample similarity:

$$\text{sim}(z_i, z_k) \rightarrow 0 \text{ for } i \neq k.$$

Consequently, the learned embedding space naturally forms well-separated clusters, which align more closely with class boundaries even without explicit supervision.

5.2.3 Implicit Class Separability

A key advantage of contrastive learning is that it induces implicit class separability without requiring labeled data. By distinguishing between different instances and enforcing invariance within the same instance, the model learns features that are inherently more discriminative.

This can be interpreted as learning a representation function f_θ such that:

$$\begin{aligned} f_\theta(x_i) &\approx f_\theta(x_j) \text{ if } x_i \sim x_j, \\ f_\theta(x_i) &\not\approx f_\theta(x_k) \text{ if } x_i \not\sim x_k, \end{aligned}$$

where similarity is defined based on augmentation invariance.

Although this does not explicitly use class labels, it often aligns well with class structure, especially in domains where samples from the same class share common underlying patterns.

5.2.4 Robustness to Observational Variability

Astronomical light curves are subject to significant variability due to:

- Measurement noise,
- Irregular sampling,
- Partial observations,
- Instrumental and environmental factors.

In an autoencoder framework, these variations are treated as part of the signal and are therefore reconstructed, leading to representations that may encode noise and irrelevant variations.

In contrast, the augmentation strategy in SimCLR explicitly introduces such variations during training. As a result, the model is trained to treat these factors as nuisance transformations rather than meaningful features.

Formally, for an augmentation $\mathcal{T}$, the model learns:

$$f_{\theta}(x) \approx f_{\theta}(\mathcal{T}(x)),$$

which enforces invariance to observational distortions.

This leads to representations that:

- Focus on underlying astrophysical structure,
- Are robust to noise and missing data,
- Generalize better to real-world conditions.

5.2.5 Improved Generalization Under Limited Labels

Another key advantage of contrastive learning is its ability to improve generalization performance in label-scarce scenarios. Since the encoder is pretrained on large

amounts of unlabeled data, it learns rich representations before encountering any labels.

During fine-tuning:

- The model requires fewer labeled examples to achieve high performance,
- The learned features already capture meaningful structure,
- The classifier can focus on refining decision boundaries rather than learning representations from scratch.

In contrast, autoencoders do not provide the same level of discriminative pretraining, resulting in weaker performance when labeled data is limited.

5.2.6 Representation Geometry and Clustering Behavior

From a geometric perspective, the difference between the two approaches can be summarized as:

- Autoencoder representations:
 - Smooth and continuous,
 - Organized by reconstruction similarity,
 - Often overlapping across classes.
- Contrastive representations:
 - Clustered and structured,
 - Organized by instance-level discrimination,
 - More aligned with class boundaries.

This difference directly explains the improved classification performance observed in Section 5.1.

5.3 Robustness Under Limited Label Conditions

A key advantage of the proposed contrastive self-supervised learning framework lies in its ability to achieve strong performance under limited labeled data conditions, which is a defining constraint in astronomical transient classification. Unlike traditional supervised approaches that rely heavily on large annotated datasets, the proposed method leverages unlabeled data for representation learning, thereby significantly reducing dependence on labeled samples.

5.3.1 Representation Learning Before Supervision

In the proposed framework, the encoder f_θ is first trained on the unlabeled dataset $\mathcal{D}_u$ using contrastive learning. During this phase, the model learns to extract meaningful features by optimizing the NT-Xent loss, without requiring any class labels.

Formally, the encoder learns a mapping:

$$f_\theta: x \rightarrow h,$$

such that the representation h captures the intrinsic structure of the data.

This pretraining process enables the model to learn:

- Temporal patterns across light curves,
- Spectral relationships across photometric bands,
- Invariance to noise and observational distortions.

As a result, the learned representations encode domain-level knowledge prior to any supervised training.

5.3.2 Efficient Use of Limited Labeled Data

Once pretraining is complete, the model is fine-tuned on a small labeled dataset:

$$\mathcal{D}_l = \{(x_i, y_i)\}.$$

At this stage, the classification head g_ψ operates on pretrained representations:

$$\hat{y}_i = g_\psi(f_\theta(x_i)).$$

Because the encoder has already learned a well-structured representation space, the supervised learning task becomes significantly simpler:

- The model does not need to learn features from scratch,
- The classifier only needs to adjust decision boundaries,
- Fewer labeled examples are sufficient for effective training.

This leads to improved performance even when:

$$|\mathcal{D}_l| \ll |\mathcal{D}_u|.$$

5.3.3 Comparison with Fully Supervised Learning

In a fully supervised setting, the model must simultaneously learn both:

- Feature representations, and
- Classification boundaries.

Under limited labeled data, this results in:

- Overfitting to noise,
- Poor generalization,
- Weak performance on rare classes.

In contrast, the proposed framework separates these two tasks:

- Pretraining phase $\rightarrow$ representation learning,
- Fine-tuning phase $\rightarrow$ classification learning.

This separation improves learning efficiency and stability.

5.3.4 Generalization and Transferability

The pretrained encoder develops representations that are:

- Generalizable across samples,
- Robust to variations in data,
- Transferable to downstream tasks.

This is particularly important in astronomy, where new transient events may differ significantly from training samples. The ability to generalize ensures that the model can:

- Adapt to unseen data,
- Maintain performance across different observational conditions,

- Detect rare and previously unseen events.
-

5.3.5 Impact on Rare Event Classification

The robustness under limited labels is especially beneficial for rare transient detection, where labeled examples are scarce. By learning a well-structured representation space, the model ensures that:

- Rare classes are not ignored due to insufficient data,
- Subtle patterns distinguishing rare events are preserved,
- Classification performance remains stable even with small labeled subsets.

This is reflected in the improved recall and F1-score observed in Section 5.1.

5.3.6 Practical Significance in Astronomy

In real-world astronomical applications, acquiring labeled data involves:

- Spectroscopic follow-up observations,
- Allocation of limited telescope time,
- Expert analysis by astronomers.

These processes are both expensive and time-consuming, making large-scale labeling impractical.

The proposed framework addresses this limitation by:

- Maximizing the utility of unlabeled data,
- Reducing reliance on expensive labeling processes,
- Enabling scalable classification for large survey datasets

5.4 Rare Event Detection Performance

The improvement in recall and F1-score achieved by the proposed contrastive learning framework has particularly significant implications for rare astronomical transient detection, which is one of the primary objectives of modern time-domain surveys. In such applications, the cost of misclassification is highly asymmetric: false negatives (missed rare events) are far more detrimental than false positives, as they result in lost opportunities for scientific discovery that cannot be recovered retrospectively.

5.4.1 Importance of Recall in Rare Event Detection

In highly imbalanced datasets, overall accuracy is often an inadequate metric for evaluating performance. A model may achieve high accuracy by correctly classifying majority classes while failing to detect rare but scientifically valuable events.

The improved recall observed in the proposed model:

$$\text{Recall}_{\text{SimCLR}} = 0.81 \text{ vs } \text{Recall}_{\text{Autoencoder}} = 0.68,$$

indicates a significantly enhanced ability to correctly identify true positive instances of rare transient classes.

From a scientific perspective, this improvement implies that:

- A larger fraction of rare events are successfully detected,
- Fewer astrophysically significant phenomena are overlooked,
- The overall scientific yield of observational surveys is increased.

Even if this improvement comes with a modest trade-off in precision, it is generally acceptable in astronomy, where follow-up observations can be used to filter false positives, but missed events cannot be recovered.

5.4.2 Balanced Performance Through F1-Score

The increase in F1-score further reinforces the model's effectiveness:

$$F1_{\text{SimCLR}} = 0.80 \text{ vs } F1_{\text{Autoencoder}} = 0.67.$$

The F1-score captures the balance between precision and recall, and its improvement indicates that the model not only detects more rare events but also maintains reasonable prediction reliability. This balanced performance is essential for deploying classification systems in real-world survey pipelines.

5.4.3 Confusion Matrix Analysis

A deeper understanding of model behavior is obtained through analysis of the confusion matrix, which reveals patterns of misclassification across different transient classes.

The primary sources of classification error are concentrated among astrophysically similar class pairs, including:

- Type Ia vs Type II Supernovae (SN Ia vs SN II)
These classes exhibit similar photometric evolution patterns, particularly in their rise and decay timescales, making them inherently difficult to distinguish using light curve data alone.
- Active Galactic Nuclei vs Quasi-Stellar Objects (AGN vs QSO)
These classes are closely related physical phenomena, differing primarily in luminosity and observational perspective rather than fundamental structure.

5.4.4 Scientific Interpretation of Errors

The observed confusion patterns are scientifically meaningful and expected, rather than indicative of model failure. This suggests that:

- The model is learning physically relevant features,
- Misclassifications occur in regions where even domain experts may encounter ambiguity,
- The model is not overfitting to noise or irrelevant patterns.

In contrast, if errors were randomly distributed across unrelated classes, it would indicate poor representation learning. The structured nature of the confusion matrix therefore provides strong evidence that the model has learned a semantically coherent representation space.

5.4.5 Implications for Real-World Deployment

The ability of the model to prioritize rare event detection and produce scientifically interpretable errors makes it highly suitable for deployment in real-world astronomical pipelines. Specifically, the model can be used to:

- Flag high-priority candidates for follow-up observation,
- Assist astronomers in triaging large volumes of transient alerts,
- Improve the efficiency of observational resource allocation.

By focusing on recall and meaningful classification behavior, the proposed system aligns closely with the operational goals of modern survey systems

5.5 Embedding Space Quality

The quality of the learned representation space is further evaluated through **dimensionality reduction and visualization**, providing qualitative insight into the structure of the embeddings produced by the proposed contrastive learning framework. In this study, **Uniform Manifold Approximation and Projection (UMAP)** is employed to project high-dimensional latent representations into a two-dimensional space for visualization.

5.5.1 Visualization of Learned Representations

Let the encoder produce latent representations:

$$h_i = f_\theta(x_i), h_i \in \mathbb{R}^d.$$

UMAP is used to obtain a low-dimensional embedding:

$$\tilde{h}_i \in \mathbb{R}^2,$$

which preserves the **local and global structure** of the original representation space.

The resulting visualization reveals that:

- Samples belonging to the same transient class form **compact and cohesive clusters**,
- Distinct classes are separated by **clear and well-defined boundaries**,
- The overall embedding space exhibits strong **cluster separability**.

5.5.2 Interpretation of Cluster Structure

The observed clustering behavior provides strong evidence that the contrastive learning framework has successfully learned **discriminative representations**. Specifically:

- **Intra-class compactness** is achieved, indicating that samples from the same class share similar embeddings,
- **Inter-class separation** is maintained, facilitating effective classification.

This structure directly aligns with the objectives of contrastive learning, which explicitly encourages:

$$\begin{aligned} \text{sim}(h_i, h_j) &\rightarrow 1 \text{ if } y_i = y_j, \\ \text{sim}(h_i, h_k) &\rightarrow 0 \text{ if } y_i \neq y_k. \end{aligned}$$

5.5.3 Partial Overlap and Scientific Validity

While most classes exhibit clear separation, partial overlap is observed among **astrophysically related classes**, particularly:

- **AGN and QSO**,
- Certain supernova subclasses (e.g., SN Ia vs SN II).

Importantly, this overlap is not indicative of model failure but rather reflects **genuine physical similarities** between these classes. These phenomena often share similar photometric characteristics, making them inherently difficult to distinguish, even for domain experts.

The presence of structured overlap suggests that:

- The model has learned **physically meaningful relationships**,
- The embedding space reflects **true astrophysical similarities**,
- The representations are not artificially forced into unrealistic separations.

5.5.4 Consistency with Quantitative Results

The qualitative observations from the UMAP visualization are consistent with the quantitative performance metrics presented in Section 5.1. Specifically:

- High classification accuracy corresponds to **well-separated clusters**,
- Improved recall and F1-score align with **clear class boundaries**,
- Structured overlap explains **specific confusion patterns** observed in classification results.

This consistency between qualitative and quantitative evaluation strengthens the validity of the proposed approach.

5.5.5 Interpretability and Model Trustworthiness

An additional advantage of embedding visualization is its contribution to **model interpretability**. By providing a visual representation of the learned feature space, UMAP enables:

- Identification of class relationships,
- Detection of ambiguous or borderline samples,
- Validation of learned representations against domain knowledge.

This interpretability is particularly important in scientific applications, where model decisions must be **understandable and justifiable**.

5.6 Limitations

While the experimental results demonstrate the effectiveness of the proposed contrastive learning framework, it is important to acknowledge several limitations that provide opportunities for future improvement and research.

5.6.1 Limited Baseline Comparison

The current evaluation compares the proposed method primarily against an **autoencoder-based baseline**. While this provides valuable insight into the advantages of contrastive learning over reconstruction-based approaches, it does not fully capture the relative performance of the model in comparison to other state-of-the-art methods. Future work should include benchmarking against:

- Fully supervised deep learning models (e.g., CNNs, RNNs),
- Transformer-based architectures,
- Hybrid and ensemble methods.

Such comparisons would provide a more comprehensive evaluation of the proposed framework.

5.6.2 Challenges with Extreme Temporal Irregularity

Astronomical light curves often exhibit **extreme irregularity and long observational gaps**, which remain challenging for sequence-based models. While the preprocessing pipeline and augmentation strategies partially address these issues, they do not fully resolve them.

Potential future directions include:

- Continuous-time models such as **Neural Ordinary Differential Equations (Neural ODEs)**,
- Advanced interpolation techniques,
- Time-aware attention mechanisms.

These approaches may provide more accurate modeling of irregular temporal dynamics.

5.6.3 Sensitivity to Augmentation Strategy

The performance of contrastive learning is highly dependent on the design of the **data augmentation strategy**. In this study, domain-specific augmentations are carefully designed to preserve astrophysical meaning while introducing variability.

However:

- The choice of augmentation types and parameters can significantly impact representation quality,
- Suboptimal augmentation policies may lead to degraded performance,
- There is no universally optimal augmentation strategy for astronomical data.

This highlights the need for systematic **ablation studies** to identify the most effective augmentation configurations.

5.6.4 Scalability and Computational Cost

Contrastive learning methods, particularly SimCLR, often require:

- Large batch sizes,
- Significant computational resources,

- Extended training time.

While these requirements are manageable in controlled experimental settings, they may pose challenges for large-scale deployment.

Future work may explore:

- Memory-efficient contrastive methods (e.g., MoCo),
 - Distributed training strategies,
 - Model optimization techniques.
-

5.6.5 Limited Exploration of Rare Extreme Events

Although the model demonstrates strong performance on the selected dataset, extremely rare and atypical transient events may still be underrepresented. The ability to generalize to such edge cases remains an open challenge.

6. Conclusion

This research presented a comprehensive and robust framework for rare astronomical transient classification based on self-supervised contrastive representation learning applied to multi-band photometric light curve data. The study addresses fundamental challenges in modern time-domain astronomy, including label scarcity, class imbalance, and irregular observational data, by leveraging the strengths of contrastive learning to extract meaningful and discriminative representations from unlabeled data.

The proposed methodology integrates several key innovations. First, a domain-aware data augmentation strategy was developed to simulate realistic observational variations while preserving astrophysical semantics, enabling effective contrastive learning in the time-series domain. Second, a hybrid convolutional-recurrent encoder architecture was designed to capture both local temporal morphology and long-range sequential dependencies, providing a rich multi-scale representation of transient events. Third, the adoption of a SimCLR-based contrastive optimization framework ensured the learning of structured and discriminative embedding spaces that naturally support classification. Finally, a two-stage training strategy, consisting of self-supervised pretraining followed by supervised fine-tuning, enabled efficient utilization of limited labeled data.

Experimental results demonstrate the effectiveness of the proposed approach, achieving 83% classification accuracy on a challenging multi-class transient dataset and significantly outperforming an autoencoder-based baseline by a substantial margin. Consistent improvements across precision, recall, and F1-score further validate the robustness of the framework, particularly in the context of rare event detection, where recall is of paramount importance. Qualitative analysis of the embedding space confirms that the model learns well-structured, semantically meaningful representations, with clustering patterns that align with known astrophysical relationships.

The findings of this study strongly support the central hypothesis that contrastive self-supervised learning provides a superior alternative to reconstruction-based approaches for representation learning in classification tasks. By explicitly enforcing discriminative structure in the latent space, the proposed framework enables more effective separation of transient classes, even under conditions of limited labeled data.

From a practical perspective, this work has significant implications for the future of astronomical data analysis. With next-generation survey facilities such as the Vera C. Rubin Observatory expected to generate massive volumes of transient alert data, scalable and label-efficient classification systems will be essential for prioritizing scientifically valuable events. The proposed framework offers a viable solution for addressing this challenge, enabling improved detection and characterization of rare astrophysical phenomena.

Beyond astronomy, the contributions of this research extend to a broader class of scientific and real-world problems characterized by abundant unlabeled data, limited

labeled samples, noisy observations, and class imbalance. The methodological framework developed in this study provides a generalizable approach for applying contrastive self-supervised learning to complex time-series classification tasks across domains such as healthcare, environmental monitoring, and financial analysis.

In conclusion, this research demonstrates that contrastive self-supervised learning is a powerful and scalable paradigm for scientific data analysis, offering substantial advantages in representation quality, classification performance, and label efficiency. Future work may further enhance this framework through improved augmentation strategies, advanced temporal modeling techniques, and integration with real-time data processing systems, paving the way for next-generation intelligent scientific discovery systems.

7.Future Work

7.1 Comprehensive Model Benchmarking

Future work should extend the evaluation to include a broader range of state-of-the-art models. This includes:

- Fully supervised recurrent architectures such as LSTM and GRU-based networks,
- Transformer-based sequence models designed for time-series data,
- Alternative self-supervised frameworks such as BYOL, MoCo, and SimSiam.

Such benchmarking would provide a more comprehensive assessment of the relative strengths and limitations of contrastive learning approaches.

7.2 Multi-Survey Data Integration and Generalization

The current study focuses on data representative of a single survey environment. Future research should explore the integration of larger and more heterogeneous datasets from multiple surveys, such as ZTF, Pan-STARRS, and LSST.

This would enable:

- Evaluation under diverse observational conditions,
- Improved robustness to instrumental and cadence variations,
- Development of models capable of cross-survey generalization.

7.3 Systematic Ablation Studies

The performance of the proposed framework is influenced by several design choices, particularly in:

- Data augmentation strategies,
- Encoder architecture components (CNN vs GRU),
- Hyperparameter configurations.

Future work should conduct detailed ablation studies to quantify the contribution of each component, providing deeper insight into the factors driving performance improvements.

7.4 Real-Time Transient Detection and Streaming Systems

With next-generation sky surveys producing continuous streams of transient alerts, extending the framework to real-time applications is a critical direction.

This includes:

- Real-time transient alert prioritization,
- Integration with streaming data pipelines,
- Development of low-latency inference systems,
- Exploration of online and continual learning approaches.

Such advancements would enable deployment in operational astronomical environments.

7.5 Multi-Modal Contrastive Learning

Astronomical data is inherently multi-modal, incorporating:

- Photometric time-series data,
- Spectroscopic observations,
- Contextual metadata (e.g., redshift, host galaxy properties).

Future research can explore multi-modal contrastive learning frameworks that jointly learn from these diverse data sources, potentially improving classification performance by leveraging complementary information.

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